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1 Complex Numbers and Plane Topology

1.1 Complex Number System

Definition 1.1

The complex numbers $\mathbb{C}$ is $\mathbb{R}^2$ with the following operations:

1. $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$
2. $(x_1, y_1) \cdot (x_2, y_2) = (x_1 x_2 - y_1 y_2, x_1 y_2 + x_2 y_1)$

Theorem 1.1

$\mathbb{C}$ is a field with $0 = (0, 0)$, $-(x, y) = (-x, -y)$, $1 = (1, 0)$ and

$$(x, y)^{-1} = \left(\frac{x}{x^2 + y^2}, \frac{-y}{x^2 + y^2} \right)$$

Notation

If $z = (x, y)$, we call x the “real part” of z and write $x = \operatorname{Re}(z)$. Call y the imaginary part of z and write $y = \operatorname{Im}(z)$.

Definition 1.2

$i = (0, 1) \in \mathbb{C}$. For $z \in \mathbb{C}$, we have $z = \operatorname{Re}(z) + \operatorname{Im}(z) \cdot i$.

Example

$$i^2 = -1 \text{ and } (-i)^2 = (0, -1)^2 = -1.$$

Definition 1.3

Conjugation

Define conjugation to be the map $\bar{\cdot} : \mathbb{C} \rightarrow \mathbb{C}$ by $x = x + iy \mapsto x - iy$.

Note

$$\operatorname{Re}(z) = \frac{z + \bar{z}}{2} \text{ and } \operatorname{Im}(z) = \frac{z - \bar{z}}{2i} = \frac{2yi}{2i} = y.$$

Definition 1.4

Polar Representation

If $z = x + iy \in \mathbb{C}$, it has a polar representation $(x, y) \rightsquigarrow (r, \theta)$ with $r = \sqrt{x^2 + y^2}$, $r \cos \theta = x$ and $r \sin \theta = y$.

Note

If z has polar coordinates (r, θ) , then $z = r \cos \theta + ir \sin \theta$.

Remark

If z_1, z_2 has polar representation $z_1 : (r_1, \theta_1)$ and $z_2 : (r_2, \theta_2)$, then

$$\begin{aligned} z_1 \cdot z_2 &= (r_1 \cos \theta_1 + ir_1 \sin \theta_1) \cdot (r_2 \cos \theta_2 + ir_2 \sin \theta_2) \\ &= r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)) \end{aligned}$$

1.2 Geometry and Topology of $\mathbb{C}$ **Note**

$\mathbb{C}$ is a metric space with $d : \mathbb{C} \times \mathbb{C} \rightarrow [0, \infty)$ given by $d(x, y) = |x - y|$, where $|\cdot| : \mathbb{C} \rightarrow [0, \infty)$ is defined to be the 2-norm, so the triangle inequality holds.

Remark

$\mathbb{C}$ is a topological space, so topological properties hold.

Definition 1.5

If z has polar representation (r, θ) , we call θ the argument of z . Write $\theta = \arg(z)$. We'll think about $\arg$ as a multi-valued function. We write $\text{Arg}(z) = \theta$ if $\theta \in (-\pi, \pi]$, and call it the principal value.

Example

If $z = r(\cos \theta + i \sin \theta)$, then we can find exactly n n th roots of z .

2 Complex Differentiation and Elementary Functions**2.1 Differentiability****Definition 2.1**

Suppose $f : U \rightarrow \mathbb{C}$ is such that $U \subseteq \mathbb{C}$ is an open set with $z_0 \in U$. We say that f has derivative $f'(z_0)$ at z_0 if

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

We say that f is differentiable at z_0 if the limit above exists.

Theorem 2.1

If f and g are differentiable at z_0 , then so is $(f \cdot g)$ and

$$(f \cdot g)' = f'(z_0)g(z_0) + f(z_0)g'(z_0)$$

2.2 Cauchy-Riemann Equations and Holomorphic Functions

Remark

If $f : \mathbb{C} \rightarrow \mathbb{C}$, we may write $f(z) = u(z) + iv(z)$ where $u, v : \mathbb{C} \rightarrow \mathbb{R}$. If f is differentiable at $z_0 = x_0 + iy_0$, say $f'(z_0) = a + ib$, then if we let $z = x + iy$ and $z_0 = x_0 + iy_0$, we have

$$\begin{aligned} R(z) &= f(z) - f(z_0) - f'(z_0)(z - z_0) \\ &= (u(z) - u(z_0) - a(x - x_0) + b(y - y_0)) + \\ &\quad i(v(z) - v(z_0) - a(y - y_0) - b(x - x_0)) \\ &= R_1(z) + iR_2(z) \end{aligned}$$

Then, since f is differentiable, we have

$$0 = \lim_{z \rightarrow z_0} \frac{R(z)}{z - z_0} = \lim_{z \rightarrow z_0} \frac{R_1(z) + iR_2(z)}{z - z_0}$$

So

$$\lim_{z \rightarrow z_0} \frac{R_1(z)}{z - z_0} = \lim_{z \rightarrow z_0} \frac{R_2(z)}{z - z_0} = 0$$

Hence, u and v are differentiable, so

$$\frac{\partial u}{\partial x}(z_0) = a, \frac{\partial v}{\partial x}(z_0) = b, \frac{\partial u}{\partial y}(z_0) = -b, \frac{\partial v}{\partial y}(z_0) = a$$

Theorem 2.2

Cauchy-Riemann Equations

If $f(z) = u(z) + iv(z)$, then f is differentiable (as a complex function) at z_0 if and only if both u and v are differentiable at z_0 and

$$\frac{\partial u}{\partial x}(z_0) = \frac{\partial v}{\partial y}(z_0)$$

and

$$\frac{\partial v}{\partial x}(z_0) = -\frac{\partial u}{\partial y}(z_0)$$

We call these the Cauchy-Riemann Equations.

Recall

Let $u : \mathbb{R}^2 \rightarrow \mathbb{R}$. Then u is differentiable at (x_0, y_0) if $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}$ exist in a neighbourhood of (x_0, y_0) and are continuous at (x_0, y_0) .

Definition 2.2

The exponential function is the function $f(x + iy) = e^x(\cos y + i \sin y)$.

$$f'(x + iy) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial y} = e^x(\cos y + i \sin y) = f(x + iy)$$

For $z \in \mathbb{C}$, we write e^z to denote $f(z)$.

Definition 2.3

If we have a function $f : U \rightarrow \mathbb{C}$ (with $U \subseteq \mathbb{C}$ open) such that f is differentiable at each point in U , then f is *holomorphic* or *analytic*.

Definition 2.4

$f : \mathbb{C} \rightarrow \mathbb{C}$ is *entire* if it is holomorphic on all of $\mathbb{C}$.

2.3 Exponential and Logarithm**Remark**

For $y \in \mathbb{R}$, we have $e^{iy} = \cos y + i \sin y$. One can see this by expanding the Taylor series for e^{iy} into the Taylor series of $\cos(y)$ and $\sin(y)$.

Definition 2.5**Logarithm**

A complex number w is a logarithm of $z \in \mathbb{C}$ if $e^w = z$.

Remark

The range of the exponential function is $\mathbb{C}^* := \mathbb{C} \setminus \{0\}$. So there is no logarithm of 0.

Definition 2.6

If $G \subseteq \mathbb{C}$ is a non-empty open set not containing 0, then a *branch of log* is a continuous function $\ell : G \rightarrow \mathbb{C}$ such that for all $z \in G$, $\ell(z)$ is a logarithm of z .

Note

A branch of log may or may not exist depending on the shape of G .

Example

$G = \mathbb{C} \setminus (-\infty, 0]$. $e^{x+iy} = e^x(\cos y + i \sin y)$. Set $l(z) = \ln(|z|) + i \operatorname{Arg}(z)$. Then

$$\begin{aligned}
 e^{l(z)} &= e^{\ln(|z|)}(\cos(\text{Arg}(z)) + i \sin(\text{Arg}(z))) \\
 &= |z|(\cos(\text{Arg}(z)) + i \sin \text{Arg}(z)) \\
 &= z
 \end{aligned}$$

This is the *principal branch of log*, denoted by $\text{Log}(z)$.

Remark

1. If we defined $l(z) = \text{Log}(x + iy) + i2\pi k$ where $k \in \mathbb{Z}$, then we get a branch of $\log$.
2. It is not possible to find a branch of $\log$ on $\mathbb{C}^*$.

Exercise 2.3.1

In polar form, the Cauchy-Riemann equations become: if $f(z) = u(z) + iv(z)$, then $ru_r = v_\theta$ and $u_\theta = -rv_r$.

Theorem 2.3

If $\ell : G \rightarrow \mathbb{C}$ is a branch of $\log$, then ℓ is holomorphic and $\ell'(z) = \frac{1}{z}$.

Proof: If ℓ is a branch of $\log$, $\ell(z) = u(z) + iv(z)$. $u(z) = \ln(|z|)$ and $v(z) = \arg(z)$. See that Cauchy-Riemann equations are satisfied. Therefore, ℓ is holomorphic and

$$\begin{aligned}
 \ell'(z) &= u_x + v_x \\
 &= u_r r_x + u_\theta + \theta_x + i(v_r r_x + v_\theta \theta_x) \\
 &= \frac{1}{r}(2x) \frac{1}{r} \left(\frac{1}{z}\right) + i \left(\frac{-y}{r^2}\right) \\
 &= \dots = \frac{1}{z}
 \end{aligned}$$

□

Theorem 2.4

Let $P \in \mathbb{C}[x, y]$ and define $F : \mathbb{C} \rightarrow \mathbb{C}$, $F(z) = P(\text{Re}(z), \text{Im}(z))$. Then F is holomorphic if and only if there exists $g \in \mathbb{C}[z]$ such that $P(x, y) = g(x + iy)$ for all $(x, y) \in \mathbb{R}^2$.

Proof: Write $f(x, y) = u(x, y) + iv(x, y)$, $f_x = u_x + iv_x$, $f_y = u_y + iv_y$. Cauchy-Riemann: $if_x = f_y$. If we write $f(x, y) = \sum_{k=0}^K f_k(x, y)$, where f_k is the terms where the sum of the exponents is k .

Claim: f satisfies Cauchy-Riemann iff each f_k does.

So, we assume $f(x, y) = c_0 x^n + c_1 x^{n-1} y + \dots c_n y^n$ for $c_0, c_1, \dots, c_n \in \mathbb{C}$. Then by CR, $c_k = i^k(n, k)c_0$. So $f(x, y) = c_0(x + iy)^n = c_0 z^n$.

□

3 Series and Power Series

3.1 Series of Numbers and Functions

Definition 3.1

Series

A *series* is a formal sum

$$\sum_{n=0}^{\infty} c_n$$

where the terms $c_n \in \mathbb{C}$. Define $S_N = \sum_{k=0}^N c_k \in \mathbb{C}$ to be the N th partial sum of the series. We say a series converges to S if $(S_N)_{N=1}^{\infty}$ converges to S . If it does not converge, then it diverges.

Lemma 3.1

If $\sum_{n=0}^{\infty} c_n$ converges, then $\lim_{n \rightarrow \infty} c_n = 0$.

Proof:

$$0 = \lim_{N \rightarrow \infty} (S_N - S_{N-1}) = \lim_{n \rightarrow \infty} c_n$$

□

Definition 3.2

A series $\sum_{n=0}^{\infty} c_n$ converges absolutely if $\sum_{n=0}^{\infty} |c_n| < \infty$.

Lemma 3.2

If $\sum_{n=0}^{\infty} c_n$ converges absolutely, then $\sum_{n=0}^{\infty} c_n$ converges.

Proof: Suppose $N < M$.

$$|S_M - S_N| = \left| \sum_{n=N+1}^M c_n \right| \leq \sum_{n=N+1}^M |c_n| \leq \sum_{n=N+1}^{\infty} |c_n| \xrightarrow{N \rightarrow \infty} 0$$

Therefore, $(S_N)_{N=1}^{\infty}$ is Cauchy. Since $\mathbb{C}$ is complete, then $(S_N)_{N=1}^{\infty}$ converges. □

Note

$\left| \sum_{n=0}^N c_n \right| \leq \sum_{n=0}^N |c_n|$, so if $\sum_{n=0}^{\infty} c_n$ converges absolutely, then

$$\left| \sum_{n=0}^{\infty} c_n \right| \leq \sum_{n=0}^{\infty} |c_n| \quad (\text{by continuity of } |\cdot|)$$

Example

For $c \in \mathbb{C}$, $\sum_{n=0}^{\infty} c^n$ converges to $\frac{1}{1-c}$ if and only if $|c| < 1$.

Definition 3.3

Suppose $(f_n)_{n=1}^{\infty}$ is a sequence of $\mathbb{C}$ -valued functions on $G \subseteq \mathbb{C}$.

1. $(f_n)_{n=1}^{\infty}$ converges pointwise to f on $S \subseteq G$ if $\lim_{n \rightarrow \infty} f_n(z) = f(z)$ for all $z \in S$.
2. $(f_n)_{n=1}^{\infty}$ converges uniformly to f on S if

$$\sup_{z \in S} |f_n(z) - f(z)| \xrightarrow{n \rightarrow \infty} 0$$

3. $(f_n)_{n=1}^{\infty}$ converges *locally uniformly* to f on G if for each $z \in G$, there is some open ball B with center z such that $(f_n)_{n=1}^{\infty}$ converges uniformly to f on B .

Example

$f_n(z) = z^n$, $G = \{z \in \mathbb{C} \mid |z| < 1\}$. Then $(f_n)_{n=0}^{\infty}$ converges locally uniformly to 0 on G , but does not converge uniformly on G .

Definition 3.4

If $(f_n)_{n=1}^{\infty}$ is a sequence of functions, $\sum_{n=0}^{\infty} f_n(z)$ is the associated series of functions. The series converges to f if the partial sums do.

We say that a series *represents* a function $g : G \rightarrow \mathbb{C}$ if for each $z \in G$, the series $\sum_{n=1}^{\infty} f_n(z)$ converges to $g(z)$.

3.2 Power Series**Example**

$z_0 \in \mathbb{C}$. $(f_n)_{n=0}^{\infty}$ with $f_n(z) = a_n(z - z_0)^n$. We get the series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$, which we call the *power series centered at z_0* .

Proposition 3.3

If $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ converges, then for $c \in \mathbb{C}$ with $|c - z_0| < |z - z_0|$, we have that $\sum_{n=0}^{\infty} a_n(c - z_0)^n$ converges absolutely.

Proof: WLOG, suppose $z_0 = 0$. Then $\sum_{n=0}^{\infty} a_n z^n$ converges. Since $\sum_{n=0}^{\infty} a_n z^n$ converges, then $a_n z^n \rightarrow 0$. Then $M := \max\{|a_n z^n| \mid n \geq 0\} < \infty$. Let $c \in \mathbb{C}$ with $|c| < |z|$. Then

$$\begin{aligned}
\sum_{n=0}^{\infty} |a_n c^n| &= \sum_{n=0}^{\infty} \left| a_n z^n \frac{c^n}{z^n} \right| \\
&= \sum_{n=0}^{\infty} |a_n z^n| \left| \frac{c}{z} \right|^n \\
&\leq \sum_{n=0}^{\infty} M \left| \frac{c}{z} \right|^n (*) \\
&< \infty
\end{aligned}$$

since $(*)$ is a geometric series in $\mathbb{R}$ with $\left| \frac{c}{z} \right| < 1$. So $\sum_{n=0}^{\infty} a_n c^n$ converges absolutely by the comparison test in $\mathbb{R}$. $\square$

Corollary 3.4

The region of convergence of a power series centered at $z_0 \in \mathbb{C}$ is a disc around z_0 (with the boundary points possibly included).

Definition 3.5

Radius of Convergence

Given $\sum_{n=0}^{\infty} a_n (z - z_0)^n$, let

$$R = \left\{ |c - z_0| : \sum_{n=0}^{\infty} a_n (c - z_0)^n \text{ converges, where } c \in C \right\}$$

We say that the *radius of convergence* of the power series is $\sup(R)$ if R is bounded, and ∞ otherwise.

Corollary 3.5

If $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ has radius of convergence R , then for $c \in \mathbb{C}$, $\sum_{n=0}^{\infty} a_n (c - z_0)^n$ converges whenever $|c - z_0| < R$ and diverges whenever $|c - z_0| > R$.

Example

$\sum_{n=0}^{\infty} z^n$ has radius of convergence 1. This represents the function $\frac{1}{1-z}$ on $\mathbb{D} := \{z \in \mathbb{C} \mid |z| < 1\}$. The partial sums are

$$\sum_{n=0}^N z^n = \frac{1 - z^{N+1}}{1 - z}$$

This converges locally uniformly to $\frac{1}{1-z}$.

Theorem 3.6**Comparison Test**

Let $S_1 := \sum_{n=0}^{\infty} a_n(z - z_0)^n$ and $S_2 := \sum_{n=0}^{\infty} b_n(z - z_0)^n$ be power series with radius of convergence R_1 and R_2 , respectively. If $|b_n| \leq M|a_n|$ for some $M > 0$, then $R_1 \leq R_2$.

Proof: Let $c \in \mathbb{C}$ with $|c - z_0| < R_1$. So S_1 converges absolutely at $z = c$. We have

$$\begin{aligned} \sum_{n=0}^{\infty} |a_n(c - z_0)^n| &= \sum_{n=0}^{\infty} |a_n| |c - z_0|^n \\ &\geq \sum_{n=0}^{\infty} \frac{|b_n|}{M} |c - z_0|^n \quad (\text{by comparison test in } \mathbb{R}) \\ &= \frac{1}{M} \sum_{n=0}^{\infty} |b_n| |c - z_0|^n \\ &= \frac{1}{M} \sum_{n=0}^{\infty} |b_n(c - z_0)^n| \end{aligned}$$

So S_2 converges absolutely at $z = c$. Hence, S_2 converges within R_1 , so $R_2 \geq R_1$. $\square$

Theorem 3.7**Cauchy-Hadamard Theorem**

The radius of convergence for a power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ is

$$\frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}}$$

where $\frac{1}{0} := \infty$ and $\frac{1}{\infty} := 0$, and

$$\limsup_{n \rightarrow \infty} a_n := \lim_{n \rightarrow \infty} \sup_{m \geq n} a_m$$

Proof: Let

$$R = \frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}} \in [0, \infty].$$

We show that this number is the radius of convergence.

If $R < \infty$, choose any $r > R$. Then

$$\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} > \frac{1}{r},$$

so

$$|a_n| r^n > 1$$

for infinitely many indices n . Hence, for any z with $|z - z_0| = r$, the terms

$$a_n(z - z_0)^n$$

do not approach 0. Thus the power series diverges for every z with $|z - z_0| = r$. Since $r > R$ was arbitrary, the power series diverges for any z with $|z - z_0| > R$.

If $R > 0$, choose any $0 < r < R$. Then

$$\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} < \frac{1}{r},$$

so

$$|a_n|r^n < 1$$

for all but finitely many indices n . Hence there exists $M > 0$ such that

$$|a_n|r^n \leq M$$

for every $n \geq 0$. Therefore, for any z with $|z - z_0| < r$,

$$\begin{aligned} \sum_{n=0}^{\infty} |a_n| |z - z_0|^n &= \sum_{n=0}^{\infty} |a_n| r^n \left| \frac{z - z_0}{r} \right|^n \\ &\leq M \sum_{n=0}^{\infty} \left| \frac{z - z_0}{r} \right|^n \\ &< \infty. \end{aligned}$$

So the power series converges absolutely for every z with $|z - z_0| < r$. Since $r < R$ was arbitrary, the power series converges for every z with $|z - z_0| < R$.

It follows that the radius of convergence is R . □

Theorem 3.8

Termwise Differentiation of Power Series

Suppose $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ has radius of convergence $R > 0$. Let f be the function it represents. Then the radius of convergence for $\sum_{n=1}^{\infty} n a_n(z - z_0)^{n-1}$ is also R , and if g is the function it represents, then f is differentiable and $f'(z) = g(z)$ for $z \in G = \{z \in \mathbb{C} \mid |z - z_0| < R\}$.

Proof: Assume WLOG that $z_0 = 0$. Let

$$L := \limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}.$$

Then $R = \frac{1}{L}$ by Cauchy-Hadamard. For the differentiated series, write $b_m = (m+1)a_{m+1}$, so that

$$\sum_{n=1}^{\infty} n a_n z^{n-1} = \sum_{m=0}^{\infty} b_m z^m.$$

Since $(m+1)^{\frac{1}{m}} \rightarrow 1$ and shifting a sequence does not change its limsup,

$$\limsup_{m \rightarrow \infty} |b_m|^{\frac{1}{m}} = \limsup_{m \rightarrow \infty} ((m+1)|a_{m+1}|)^{\frac{1}{m}} = L.$$

Hence the differentiated series also has radius of convergence $\frac{1}{L} = R$.

Now fix $w \in \mathbb{C}$ with $|w| < R$. Choose ρ such that $|w| < \rho < R$. Then both

$$\sum_{n=0}^{\infty} a_n z^n \quad \text{and} \quad \sum_{n=1}^{\infty} n a_n z^{n-1}$$

converge uniformly on $|z| \leq \rho$ by comparison with a convergent geometric series.

For $z \neq w$ with $|z| \leq \rho$, we have

$$\frac{f(z) - f(w)}{z - w} = \sum_{n=1}^{\infty} a_n \frac{z^n - w^n}{z - w} = \sum_{n=1}^{\infty} a_n \sum_{k=0}^{n-1} z^k w^{n-1-k}.$$

This series converges uniformly for $|z| \leq \rho$, since

$$\left| a_n \sum_{k=0}^{n-1} z^k w^{n-1-k} \right| \leq n |a_n| \rho^{n-1},$$

and $\sum_{n=1}^{\infty} n |a_n| \rho^{n-1}$ converges. Hence we may pass the limit $z \rightarrow w$ through the sum:

$$\lim_{z \rightarrow w} \frac{f(z) - f(w)}{z - w} = \sum_{n=1}^{\infty} a_n n w^{n-1} = g(w).$$

Therefore f is differentiable at w and $f'(w) = g(w)$.

Since w was arbitrary with $|w| < R$, we have

$$f'(z) = \sum_{n=1}^{\infty} n a_n z^{n-1}$$

for all $z \in \{z \in \mathbb{C} \mid |z| < R\}$. □

Example

$f(z) = \sum_{n=0}^{\infty} \frac{1}{n!} z^n$, $f'(z) = f(z)$, $f(0) = 1$. Therefore, $f(z) = e^z$.

Corollary 3.9

If $f : \mathbb{C} \rightarrow \mathbb{C}$ is represented by a power series, then f is smooth.

4 Complex Integration and Cauchy Theory

4.1 Complex-Valued Integrals

Definition 4.1

Piecewise Continuity

Let I be an interval in $\mathbb{R}$ with endpoints $a < b$. A function $\varphi : I \rightarrow \mathbb{C}$ is *piecewise continuous* if there exist points

$$a = s_0 < s_1 < \dots < s_n = b$$

with $I \subseteq [a, b]$ and continuous functions $g_j : [s_{j-1}, s_j] \rightarrow \mathbb{C}$ for $j = 1, \dots, n$ such that

$$\varphi(t) = g_j(t) \quad \text{for all } t \in I \cap (s_{j-1}, s_j).$$

Definition 4.2

Path

A *path* is a function $\gamma : [a, b] \rightarrow \mathbb{C}$.

Definition 4.3

Piecewise $\mathcal{C}^1$

A path $\varphi : [a, b] \rightarrow \mathbb{C}$ is *piecewise $\mathcal{C}^1$* if there is a partition

$$a = t_0 < t_1 < \dots < t_n = b$$

such that $\varphi|_{[t_{j-1}, t_j]}$ is $\mathcal{C}^1$ for every $1 \leq j \leq n$, where derivatives at the endpoints of each subinterval are understood as one-sided derivatives. In particular, φ is continuous on $[a, b]$, while φ' may fail to exist at the interior partition points; at each such point, its one-sided limits exist.

Definition 4.4

Integral of a Complex Valued Function

Let $\varphi : [a, b] \rightarrow \mathbb{C}$ be piecewise continuous. Then, $\operatorname{Re}(\varphi)$ and $\operatorname{Im}(\varphi)$ are also piecewise continuous. Define

$$\int_a^b \varphi(t) dt := \int_a^b \operatorname{Re}(\varphi) dt + i \int_a^b \operatorname{Im}(\varphi) dt$$

Note

The integral above is well-defined because a piecewise continuous function φ has only finitely many possible discontinuities, namely at the partition points $s_0, \dots, s_n$. Moreover, each g_j is continuous on the compact interval $[s_{j-1}, s_j]$, hence bounded. Since there are only finitely many g_j 's, φ is bounded. The same applies to $\operatorname{Re}(\varphi)$ and $\operatorname{Im}(\varphi)$, so both are bounded with finitely many discontinuities, hence Riemann integrable.

Theorem 4.1 Fundamental Theorem of Calculus (Complex-Valued Functions)

Let $\varphi : [a, b] \rightarrow \mathbb{C}$ be continuous and piecewise $\mathcal{C}^1$ on $[a, b]$. Then

$$\int_a^b \varphi'(t) dt = \varphi(b) - \varphi(a)$$

Proof: Write $\varphi(t) = u(t) + iv(t)$, where $u, v : [a, b] \rightarrow \mathbb{R}$. Since φ is continuous and piecewise $\mathcal{C}^1$, the functions u and v are continuous and piecewise $\mathcal{C}^1$. Moreover,

$$\varphi'(t) = u'(t) + iv'(t)$$

wherever φ' is defined.

By the definition of the complex-valued integral and the real fundamental theorem of calculus,

$$\begin{aligned} \int_a^b \varphi'(t) dt &= \int_a^b u'(t) dt + i \int_a^b v'(t) dt \\ &= (u(b) - u(a)) + i(v(b) - v(a)) \\ &= (u(b) + iv(b)) - (u(a) + iv(a)) \\ &= \varphi(b) - \varphi(a). \end{aligned}$$

□

Theorem 4.2 Triangle Inequality for Complex-Valued Integrals

Let $\varphi : [a, b] \rightarrow \mathbb{C}$ be piecewise continuous. Then

$$\left| \int_a^b \varphi(t) dt \right| \leq \int_a^b |\varphi(t)| dt.$$

Proof: If $\int_a^b \varphi(t) dt = 0$, then the result is immediate. Otherwise, define

$$\lambda := \frac{\left| \int_a^b \varphi(t) dt \right|}{\int_a^b |\varphi(t)| dt}.$$

Then $|\lambda| = 1$ and

$$\begin{aligned}
\left| \int_a^b \varphi(t) dt \right| &= \lambda \int_a^b \varphi(t) dt \\
&= \int_a^b \lambda \varphi(t) dt \\
&= \operatorname{Re} \left(\int_a^b \lambda \varphi(t) dt \right) \\
&= \int_a^b \operatorname{Re}(\lambda \varphi(t)) dt \\
&\leq \int_a^b |\lambda \varphi(t)| dt \\
&= \int_a^b |\varphi(t)| dt.
\end{aligned}$$

□

4.2 Path Integrals

Definition 4.5

Path Integral

Let $\gamma : [0, 1] \rightarrow G$ be a piecewise- $\mathcal{C}^1$ path, and suppose $f : G \rightarrow \mathbb{C}$ is continuous on a subset of G containing the range of γ . Define

$$\int_{\gamma} f(z) dz := \int_0^1 f(\gamma(t)) \gamma'(t) dt.$$

Note

Invariance Under Reparametrization

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be piecewise- $\mathcal{C}^1$, and let $\beta : [a_1, b_1] \rightarrow [a, b]$ be increasing and piecewise- $\mathcal{C}^1$ with $\beta(a_1) = a$ and $\beta(b_1) = b$. Set $\gamma_1 = \gamma \circ \beta$. If f is continuous on the range of γ , then

$$\begin{aligned}
\int_{\gamma_1} f(z) dz &= \int_{a_1}^{b_1} f(\gamma_1(s)) \gamma_1'(s) ds \\
&= \int_{a_1}^{b_1} f(\gamma(\beta(s))) \gamma'(\beta(s)) \beta'(s) ds \\
&= \int_a^b f(\gamma(t)) \gamma'(t) dt \\
&= \int_{\gamma} f(z) dz.
\end{aligned}$$

Theorem 4.3 **Fundamental Theorem of Calculus (For Path Integrals)**

If $\gamma : [a, b] \rightarrow \mathbb{C}$ is piecewise- $\mathcal{C}^1$, and if $f : \mathbb{C} \rightarrow \mathbb{C}$ is holomorphic on an open set that contains the range of γ , then

$$\int_{\gamma} f'(z) dz = f(\gamma(b)) - f(\gamma(a))$$

Proof:

$$\begin{aligned} \int_{\gamma} f'(z) dz &= \int_a^b f'(\gamma(t)) \gamma'(t) dt \\ &= \int_a^b (f \circ \gamma)'(t) dt \\ &= (f \circ \gamma)(b) - (f \circ \gamma)(a) \\ &= f(\gamma(b)) - f(\gamma(a)). \end{aligned}$$

□

Note
Additivity Under Subdivision

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be piecewise- $\mathcal{C}^1$, let $c \in [a, b]$, and define

$$\gamma_1 = \gamma|_{[a, c]}, \quad \gamma_2 = \gamma|_{[c, b]}.$$

If f is continuous on the range of γ , then

$$\int_{\gamma} f(z) dz = \int_{\gamma_1} f(z) dz + \int_{\gamma_2} f(z) dz.$$

Example

Let $z_0 \in \mathbb{C}$, $R > 0$, and let $\gamma : [0, 2\pi] \rightarrow \mathbb{C}$ be given by $\gamma(t) = z_0 + Re^{it}$. If $n \in \mathbb{Z}$ and $n \neq -1$, then

$$\int_{\gamma} (z - z_0)^n dz = 0.$$

Indeed, $f(z) = \frac{(z - z_0)^{n+1}}{n+1}$ satisfies $f'(z) = (z - z_0)^n$, so by the fundamental theorem of calculus for path integrals,

$$\int_{\gamma} (z - z_0)^n dz = f(\gamma(2\pi)) - f(\gamma(0)) = 0.$$

For $n = -1$, we have

$$\begin{aligned}
 \int_{\gamma} \frac{1}{z - z_0} dz &= \int_0^{2\pi} \frac{1}{Re^{it}} iRe^{it} dt \\
 &= i \int_0^{2\pi} dt \\
 &= 2\pi i.
 \end{aligned}$$

Definition 4.6**Length of a Path**

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be piecewise- $\mathcal{C}^1$. The length of γ is

$$L(\gamma) := \int_a^b |\gamma'(t)| dt.$$

Theorem 4.4**ML Inequality**

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be piecewise- $\mathcal{C}^1$, and suppose f is continuous on the range of γ . If

$$M := \max_{t \in [a, b]} |f(\gamma(t))|,$$

then

$$\left| \int_{\gamma} f(z) dz \right| \leq ML(\gamma).$$

Proof:

$$\begin{aligned}
 \left| \int_{\gamma} f(z) dz \right| &= \left| \int_a^b f(\gamma(t)) \gamma'(t) dt \right| \\
 &\leq \int_a^b |f(\gamma(t)) \gamma'(t)| dt \\
 &= \int_a^b |f(\gamma(t))| |\gamma'(t)| dt \\
 &\leq \int_a^b M |\gamma'(t)| dt \\
 &= ML(\gamma).
 \end{aligned}$$

□

Theorem 4.5**Interchange of Limit and Path Integral**

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be piecewise- $\mathcal{C}^1$, and let $(f_n)_{n=1}^\infty$ be continuous on the range of γ . If $f_n \rightarrow f$ uniformly on the range of γ , then

$$\lim_{n \rightarrow \infty} \int_{\gamma} f_n(z) dz = \int_{\gamma} f(z) dz.$$

Proof: For each n , by the ML inequality applied to $f_n - f$,

$$\begin{aligned} \left| \int_{\gamma} f_n(z) dz - \int_{\gamma} f(z) dz \right| &= \left| \int_{\gamma} (f_n - f)(z) dz \right| \\ &\leq L(\gamma) \max_{t \in [a, b]} |f_n(\gamma(t)) - f(\gamma(t))|. \end{aligned}$$

The maximum on the right tends to 0 by uniform convergence on the range of γ . Hence the integrals converge to $\int_{\gamma} f(z) dz$. $\square$

4.3 Goursat's Theorem**Example**

Let $R_{a,b}$ be the rectangle with vertices $-a + ib, a + ib, a - ib, -a - ib$, oriented counter-clockwise. We want to compute

$$\int_{R_{a,b}} e^{-z^2} dz.$$

Since

$$e^{-z^2} = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{n!},$$

define

$$g(z) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)n!}.$$

Then, by termwise differentiation of power series,

$$g'(z) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{n!} = e^{-z^2}.$$

Therefore

$$\int_{R_{a,b}} e^{-z^2} dz = \int_{R_{a,b}} g'(z) dz = 0.$$

Equivalently, if $\gamma_1, \dots, \gamma_4$ are the four edges of the rectangle, then

$$\sum_{j=1}^4 \int_{\gamma_j} e^{-z^2} dz = 0.$$

One can compute the four edge integrals separately and use this identity in the Fourier transform of the Gaussian.

Theorem 4.6

Cauchy's Theorem for Triangles (Goursat's Lemma)

Let Δ be a closed solid triangle in $\mathbb{C}$, and let T be a closed path which goes once (counter-clockwise) around the boundary of Δ . If f is holomorphic on an open set containing Δ , then

$$\int_T f(z) dz = 0.$$

Proof: Let

$$I = \int_T f(z) dz.$$

Set $\Delta_0 = \Delta$ and $T_0 = T$. Split Δ_0 into four closed solid triangles by joining the midpoints of its sides, and let $S_1, \dots, S_4$ be their counter-clockwise boundary loops.

When the four boundary integrals are added, every interior edge is traversed once in each direction. Hence the integrals over the interior edges cancel, while the remaining outer edges give exactly the integral over T . Therefore

$$I = \sum_{j=1}^4 \int_{S_j} f(z) dz.$$

By the triangle inequality, at least one of these four integrals has absolute value at least $\frac{|I|}{4}$. Thus there is some $j_1 \in \{1, 2, 3, 4\}$ such that

$$\left| \int_{S_{j_1}} f(z) dz \right| \geq \frac{|I|}{4}.$$

Let Δ_1 be the corresponding subtriangle and let $T_1 = S_{j_1}$.

Now repeat the same construction inside the selected subtriangle, and then continue inductively. This gives nested closed solid triangles

$$\Delta_0 \supseteq \Delta_1 \supseteq \Delta_2 \supseteq \dots$$

with counter-clockwise boundary loops $T_0, T_1, T_2, \dots$ such that

$$\left| \int_{T_n} f(z) dz \right| \geq \frac{|I|}{4^n},$$

while

$$L(T_n) = 2^{-n} L(T), \quad \text{diam}(\Delta_n) = 2^{-n} \text{diam}(\Delta).$$

The length and diameter identities hold because each subdivision replaces the chosen triangle by a similar triangle whose side lengths are half as large.

Note that Δ is compact, and for each $n \in \mathbb{N}$, Δ_n is closed in the subspace topology of Δ (since Δ_n is closed in $\mathbb{C}$). Note further that Δ_n is non-empty with $\Delta_1 \supseteq \Delta_2 \supseteq \dots \supseteq \Delta_n$. Therefore, $\bigcap_{n=1}^{\infty} \Delta_n \neq \emptyset$, so there exists $z_0 \in \bigcap_{n=0}^{\infty} \Delta_n$. Since f is holomorphic at z_0 , for every $\varepsilon > 0$ there is $\delta > 0$ such that

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \eta(z)(z - z_0),$$

where $|\eta(z)| \leq \varepsilon$ whenever $|z - z_0| < \delta$. Choose n large enough that $\Delta_n \subseteq B_\delta(z_0)$; this is possible because $z_0 \in \Delta_n$ and $\text{diam}(\Delta_n) \rightarrow 0$.

On the loop T_n , the expansion above gives

$$\int_{T_n} f(z) dz = \int_{T_n} f(z_0) dz + \int_{T_n} f'(z_0)(z - z_0) dz + \int_{T_n} \eta(z)(z - z_0) dz.$$

The first two integrals are 0 because the functions $f(z_0)$ and $f'(z_0)(z - z_0)$ have complex primitives $f(z_0)z$ and $f'(z_0)\frac{(z - z_0)^2}{2}$, and T_n is closed. Hence

$$\int_{T_n} f(z) dz = \int_{T_n} \eta(z)(z - z_0) dz.$$

Since $|\eta(z)| \leq \varepsilon$ and $\Delta_n \subseteq B_\delta(z_0)$ By the ML inequality,

$$\begin{aligned} \left| \int_{T_n} f(z) dz \right| &= \left| \int_{T_n} \eta(z)(z - z_0) dz \right| \\ &\leq \varepsilon \text{diam}(\Delta_n) L(T_n) \\ &= \varepsilon 4^{-n} \text{diam}(\Delta) L(T). \end{aligned}$$

Here we used that every point of T_n lies in Δ_n , so $|z - z_0| \leq \text{diam}(\Delta_n)$. Combining this with

$$|I| \leq 4^n \left| \int_{T_n} f(z) dz \right|$$

gives

$$|I| \leq \varepsilon \text{diam}(\Delta) L(T).$$

Since $\varepsilon > 0$ was arbitrary, $I = 0$. □

4.4 Cauchy's Theorem and Integral Formula

Theorem 4.7

Cauchy's Theorem for Convex Regions

Suppose $f : \mathbb{C} \rightarrow \mathbb{C}$ is holomorphic in $G \subseteq \mathbb{C}$ and G is convex (if $z_1, z_2 \in G$, then $[z_1, z_2] \in G$), then for any piecewise $\mathcal{C}'$ closed curve γ in G , we have

$$\int_{\gamma} f(z) dz = 0$$

Proof: Note that if $f(z) = g'(z)$ with g holomorphic in G , then the theorem follows from the fundamental theorem of calculus (since γ is closed).

Fix $z_0 \in G$. Define

$$g(z) = \int_{[z_0, z]} f(\zeta) d\zeta.$$

This is well-defined because G is convex, so $[z_0, z] \subseteq G$ for every $z \in G$.

We claim that $g'(z) = f(z)$ for every $z \in G$. Fix $z \in G$ and take $h \in \mathbb{C}$ with $h \neq 0$ and $z + h \in G$. Since G is convex, the triangle with vertices $z_0, z, z + h$ lies in G . By Cauchy's theorem for triangles,

$$\int_{[z_0, z]} f(\zeta) d\zeta + \int_{[z, z+h]} f(\zeta) d\zeta + \int_{[z+h, z_0]} f(\zeta) d\zeta = 0.$$

Hence

$$g(z+h) - g(z) = \int_{[z, z+h]} f(\zeta) d\zeta.$$

Therefore

$$\frac{g(z+h) - g(z)}{h} - f(z) = \frac{1}{h} \int_{[z, z+h]} (f(\zeta) - f(z)) d\zeta.$$

Parametrize $[z, z+h]$ by $\zeta(t) = z + th$, $0 \leq t \leq 1$. Then

$$\left| \frac{g(z+h) - g(z)}{h} - f(z) \right| \leq \int_0^1 |f(z+th) - f(z)| dt.$$

Since f is continuous at z , the right-hand side tends to 0 as $h \rightarrow 0$. Thus $g'(z) = f(z)$.

Now the fundamental theorem of calculus for path integrals gives

$$\int_{\gamma} f(z) dz = \int_{\gamma} g'(z) dz = g(\gamma(b)) - g(\gamma(a)) = 0,$$

because γ is closed. □

Theorem 4.8

Cauchy's Formula for the Circle

Let $\bullet$ be a closed (solid) disc in $\mathbb{C}$, and C be a closed path oriented counter-clockwise around the boundary of $\bullet$. If f is holomorphic on an open set that contains $\bullet$, then if z is in the interior of $\bullet$, we have

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta$$

Proof: Let U be an open set containing $\bullet$ on which f is holomorphic. Fix $z_0 \in \text{int}(\bullet)$. Write $\bullet = \{z \in \mathbb{C} \mid |z - a| \leq R\}$, so C goes once counter-clockwise around $\{z \in \mathbb{C} \mid |z - a| = R\}$.

Define $h : U \rightarrow \mathbb{C}$ by

$$h(\zeta) = \begin{cases} \frac{f(\zeta) - f(z_0)}{\zeta - z_0}, & \text{if } \zeta \neq z_0 \\ f'(z_0), & \text{if } \zeta = z_0. \end{cases}$$

Then h is continuous on U and holomorphic on $U \setminus \{z_0\}$. By the same argument as in Goursat's lemma, the one exceptional point z_0 contributes nothing, so

$$\int_C h(\zeta) d\zeta = 0.$$

Hence

$$\int_C \frac{f(\zeta)}{\zeta - z_0} d\zeta = f(z_0) \int_C \frac{1}{\zeta - z_0} d\zeta.$$

It remains to compute the last integral. By invariance under reparametrization, we may use the standard counter-clockwise parametrization $\gamma(t) = a + Re^{it}$, $0 \leq t \leq 2\pi$. Since $z_0 \in \text{int}(\bullet)$, $|z_0 - a| < R$. Thus

$$\begin{aligned} \int_C \frac{1}{\zeta - z_0} d\zeta &= \int_0^{2\pi} \frac{iRe^{it}}{a - z_0 + Re^{it}} dt \\ &= i \int_0^{2\pi} \frac{1}{1 + \frac{a - z_0}{Re^{it}}} dt \\ &= i \int_0^{2\pi} \sum_{n=0}^{\infty} \left(-\frac{a - z_0}{R} \right)^n e^{-int} dt \\ &= 2\pi i. \end{aligned}$$

The geometric series above converges uniformly in t because $\frac{|a - z_0|}{R} < 1$, so the interchange of the sum and integral is valid. Therefore

$$\int_C \frac{f(\zeta)}{\zeta - z_0} d\zeta = 2\pi i f(z_0),$$

which is the desired formula. □

4.5 Cauchy Integrals and Analyticity

Note

Cauchy Integral Outside the Disc

Let $\bullet = \overline{D}(a, R)$ be a closed disc, and let C be its counter-clockwise boundary. Suppose that f is holomorphic on an open set U containing $\bullet$. Fix $w \notin \bullet$ and define

$$g(\zeta) = \frac{f(\zeta)}{\zeta - w}.$$

Since w is outside the closed disc, there is positive distance between w and $\bullet$. Also, since $\bullet$ is compact and $\bullet \subseteq U$, we can choose a slightly larger concentric open disc which is still contained in U and still does not contain w . That is, there exists $\rho > R$ such that

$$\bullet \subseteq D(a, \rho) \subseteq U \quad \text{and} \quad w \notin D(a, \rho).$$

On this disc, the function $\zeta \mapsto \frac{1}{\zeta - w}$ is holomorphic. Hence $g = f \cdot \left(\zeta \mapsto \frac{1}{\zeta - w}\right)$ is holomorphic on the convex region $D(a, \rho)$. Cauchy's theorem for convex regions therefore gives

$$\int_C \frac{f(\zeta)}{\zeta - w} d\zeta = \int_C g(\zeta) d\zeta = 0.$$

Corollary 4.9

Mean Value Property for Holomorphic Functions

Suppose f is holomorphic on the disc $\{z \in \mathbb{C} \mid |z - z_0| < R\}$. Let $0 < r < R$, and let C_r be the circle of radius r centered at z_0 , oriented counter-clockwise. By Cauchy's formula,

$$f(z_0) = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{\zeta - z_0} d\zeta.$$

Parametrize C_r by $\gamma(t) = z_0 + re^{it}$ for $0 \leq t \leq 2\pi$, so that $\gamma'(t) = ire^{it}$. Then

$$\begin{aligned} f(z_0) &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(z_0 + re^{it})}{re^{it}} ire^{it} dt \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt. \end{aligned}$$

Definition 4.7

Let γ be a piecewise- $\mathcal{C}^1$ path, and let $\varphi : \text{Range}(\gamma) \rightarrow \mathbb{C}$ be continuous. The *Cauchy integral* of φ along γ is the function

$$F : \mathbb{C} \setminus \text{Range}(\gamma) \rightarrow \mathbb{C}, \quad F(z) = \int_{\gamma} \frac{\varphi(\zeta)}{\zeta - z} d\zeta.$$

Theorem 4.10

Cauchy Integrals Are Holomorphic

Let γ be a piecewise- $\mathcal{C}^1$ path and let $\varphi : \text{Range}(\gamma) \rightarrow \mathbb{C}$ be continuous. Define

$$F(z) = \int_{\gamma} \frac{\varphi(\zeta)}{\zeta - z} d\zeta, \quad z \in \mathbb{C} \setminus \text{Range}(\gamma).$$

Then F is holomorphic on $\mathbb{C} \setminus \text{Range}(\gamma)$.

More precisely, fix $z_0 \in \mathbb{C} \setminus \text{Range}(\gamma)$ and set

$$R = \text{dist}(z_0, \text{Range}(\gamma)).$$

Then, for $|z - z_0| < R$,

$$F(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n, \quad a_n = \int_{\gamma} \frac{\varphi(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

Proof: Fix z_0 not in the range of γ . Since $\text{Range}(\gamma)$ is closed and $z_0 \notin \text{Range}(\gamma)$, there is positive distance between z_0 and $\text{Range}(\gamma)$; otherwise, z_0 would be a limit point of $\text{Range}(\gamma)$ and hence would belong to it. Set

$$R = \text{dist}(z_0, \text{Range}(\gamma)) := \inf_{\zeta \in \text{Range}(\gamma)} |\zeta - z_0| > 0.$$

So,

$$\begin{aligned} \frac{1}{\zeta - z} &= \frac{1}{(\zeta - z_0) - (z - z_0)} \\ &= \frac{1}{\zeta - z_0} \left(\frac{1}{1 - \left(\frac{z - z_0}{\zeta - z_0} \right)} \right) \\ &= \frac{1}{\zeta - z_0} \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\zeta - z_0)^n} \end{aligned}$$

This, as a function of ζ , converges locally uniformly when

$$\frac{|z - z_0|}{|\zeta - z_0|} < 1$$

Therefore,

$$\frac{\varphi(\zeta)}{\zeta - z} = \frac{\varphi(\zeta)}{\zeta - z_0} \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\zeta - z_0)^n}$$

Hence,

$$\begin{aligned} \int_{\gamma} \frac{\varphi(\zeta)}{\zeta - z} d\zeta &= \sum_{n=0}^{\infty} \int_{\gamma} \frac{\varphi(\zeta)(z - z_0)^n}{(\zeta - z_0)^{n+1}} d\zeta \\ &= \sum_{n=0}^{\infty} \left(\int_{\gamma} \frac{\varphi(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta \right) (z - z_0)^n \end{aligned}$$

Therefore, F is represented by

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n$$

where

$$a_n = \int_{\gamma} \frac{\varphi(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta$$

with radius of convergence at least R .

□

Note

Holomorphic Functions Are Locally Analytic

Suppose f is holomorphic on the disc

$$D(z_0, R) = \{z \in \mathbb{C} \mid |z - z_0| < R\}.$$

Fix $r < R$, and let C_r be the positively oriented boundary of the closed disc $\overline{D}(z_0, r)$. Since $\overline{D}(z_0, r) \subseteq D(z_0, R)$, the function f is holomorphic on an open set containing this closed disc. Hence, by Cauchy's integral formula, for every $z \in D(z_0, r)$,

$$f(z) = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Thus f is a Cauchy integral on $D(z_0, r)$. By Theorem 3.10,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n, \quad a_n = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta,$$

and this power series has radius of convergence at least r . Since we may choose such an $r > 0$ around every point in the domain of a holomorphic function, every holomorphic function is locally analytic.

Moreover, we can compute $f^{(n)}$ by applying Termwise Differentiation of Power Series theorem and obtain

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta$$

Corollary 4.11

Holomorphic Functions are Smooth

If $G \subseteq \mathbb{C}$ is open and $f : G \rightarrow \mathbb{C}$ is holomorphic, then f' is holomorphic on G . In particular, every holomorphic function is infinitely differentiable.

Proof: Fix $z_0 \in G$. Since G is open, there is an $r > 0$ such that $\{z \in \mathbb{C} \mid |z - z_0| < r\} \subseteq G$. By Theorem 3.10, f has a power-series representation

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

on a neighbourhood of z_0 . Differentiating term by term gives

$$f'(z) = \sum_{n=1}^{\infty} n a_n (z - z_0)^{n-1}.$$

The differentiated series has the same radius of convergence, so it defines a holomorphic function near z_0 . Thus f' is holomorphic at every point of G , and hence on all of G .

Repeating the argument shows that all higher derivatives of f exist and are holomorphic.

□

Example

The converse is false. Let

$$f(x + iy) = \frac{1}{1 + x^2}.$$

Then f is smooth as a function $\mathbb{R}^2 \rightarrow \mathbb{C}$. However, if $f = u + iv$, then

$$u(x, y) = \frac{1}{1 + x^2}, \quad v(x, y) = 0.$$

Hence

$$u_x = -2 \frac{x}{(1 + x^2)^2}, \quad v_y = 0.$$

The Cauchy-Riemann equation $u_x = v_y$ fails whenever $x \neq 0$, so f is not holomorphic on any open subset of $\mathbb{C}$.

4.6 Morera's Theorem and Global Consequences

Corollary 4.12

Suppose

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n \quad \text{and} \quad g(z) = \sum_{n=0}^{\infty} b_n (z - z_0)^n$$

have radii of convergence R_1 and R_2 , respectively. Define

$$c_n = \sum_{k=0}^n a_k b_{n-k}.$$

Then the power series

$$\sum_{n=0}^{\infty} c_n (z - z_0)^n$$

has radius of convergence at least $\min\{R_1, R_2\}$ and represents the product $f(z)g(z)$ whenever $|z - z_0| < \min\{R_1, R_2\}$.

Proof: Fix z with $|z - z_0| < \min\{R_1, R_2\}$. Both series defining $f(z)$ and $g(z)$ converge absolutely. Hence their Cauchy product also converges absolutely, and

$$\begin{aligned} f(z)g(z) &= \left(\sum_{k=0}^{\infty} a_k (z - z_0)^k \right) \left(\sum_{j=0}^{\infty} b_j (z - z_0)^j \right) \\ &= \sum_{n=0}^{\infty} \sum_{k=0}^n a_k b_{n-k} (z - z_0)^n \\ &= \sum_{n=0}^{\infty} c_n (z - z_0)^n. \end{aligned}$$

This holds for every z satisfying $|z - z_0| < \min\{R_1, R_2\}$. Therefore, the radius of convergence of the Cauchy-product series is at least $\min\{R_1, R_2\}$, and its sum is fg on that disc. $\square$

Corollary 4.13

Converse of Goursat's Lemma

Let $G \subseteq \mathbb{C}$ be open and let $f : G \rightarrow \mathbb{C}$ be continuous. Suppose that for every closed solid triangle $\Delta \subseteq G$, oriented counter-clockwise along its boundary $\partial\Delta$, one has

$$\int_{\partial\Delta} f(z) dz = 0.$$

Then f is holomorphic on G .

Proof: Fix $z_0 \in G$. Choose $r > 0$ such that the open disc

$$D = \{z \in \mathbb{C} \mid |z - z_0| < r\}$$

is contained in G . Since D is convex, the line segment $[z_0, z]$ lies in D for every $z \in D$. Define

$$F(z) = \int_{[z_0, z]} f(\zeta) d\zeta.$$

Fix $z \in D$ and take $h \in \mathbb{C}$ such that $h \neq 0$ and $z + h \in D$. The triangle with vertices $z_0, z, z + h$ lies in D , so the hypothesis gives

$$\int_{[z_0, z]} f(\zeta) d\zeta + \int_{[z, z+h]} f(\zeta) d\zeta + \int_{[z+h, z_0]} f(\zeta) d\zeta = 0.$$

Therefore,

$$F(z + h) - F(z) = \int_{[z, z+h]} f(\zeta) d\zeta.$$

Parametrizing $[z, z + h]$ by $\zeta(t) = z + th$, $0 \leq t \leq 1$, yields

$$\frac{F(z + h) - F(z)}{h} - f(z) = \int_0^1 (f(z + th) - f(z)) dt.$$

Since f is continuous at z , the right-hand side tends to 0 as $h \rightarrow 0$. Hence $F'(z) = f(z)$ for every $z \in D$. Thus F is holomorphic on D , and Corollary 4.11 implies that $f = F'$ is holomorphic on D .

Since z_0 was arbitrary, f is holomorphic on G . □

Theorem 4.14

Liouville Theorem

Suppose $f : \mathbb{C} \rightarrow \mathbb{C}$ is entire and bounded, then f is constant.

Proof: Fix $z_0 \in \mathbb{C}$. Let C_r be the circle with center z_0 and radius r oriented counterclockwise. Then

$$f'(z_0) = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^2} d\zeta$$

So

$$\begin{aligned} |f'(z_0)| &\leq \frac{1}{2\pi} \left| \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^2} d\zeta \right| \\ &\leq \frac{1}{2\pi} L(C_r) \sup_{\zeta \in C_r} \left\{ \left| \frac{f(\zeta)}{(\zeta - z_0)^2} \right| \right\} \\ &\leq r \frac{M}{r^2} = \frac{1}{r} M \xrightarrow{r \rightarrow \infty} 0 \end{aligned}$$

□

Theorem 4.15

Fundamental Theorem of Algebra

Every nonconstant polynomial over $\mathbb{C}$ splits into linear factors. More explicitly, if

$$P(z) = \sum_{k=0}^n a_k z^k, \quad a_n \neq 0, \quad n \geq 1,$$

then there exist $b_1, \dots, b_n \in \mathbb{C}$ such that

$$P(z) = a_n \prod_{j=1}^n (z - b_j).$$

Proof: We first show that P has a zero. Suppose instead that $P(z) \neq 0$ for every $z \in \mathbb{C}$. Then

$$f : \mathbb{C} \rightarrow \mathbb{C}, \quad f(z) = \frac{1}{P(z)}$$

is entire.

Set $M = \sum_{k=0}^{n-1} |a_k|$. For $|z| \geq 1$,

$$\left| \sum_{k=0}^{n-1} a_k z^k \right| \leq \sum_{k=0}^{n-1} |a_k| |z|^k \leq M |z|^{n-1}.$$

Choose $R \geq 1$ such that $\frac{M}{R} \leq \frac{|a_n|}{2}$. If $|z| \geq R$, then

$$\begin{aligned} |P(z)| &\geq |a_n| |z|^n - \left| \sum_{k=0}^{n-1} a_k z^k \right| \\ &\geq |a_n| |z|^n - M |z|^{n-1} \\ &\geq \frac{|a_n|}{2} |z|^n. \end{aligned}$$

Consequently,

$$|f(z)| \leq \frac{2}{|a_n| |z|^n} \leq \frac{2}{|a_n| R^n}$$

whenever $|z| \geq R$.

The function f is also bounded on the compact disc $\{z \in \mathbb{C} \mid |z| \leq R\}$. Thus f is bounded on $\mathbb{C}$, so Liouville's theorem implies that f is constant. This would force P to be constant, a contradiction. Therefore, P has a zero $b_1 \in \mathbb{C}$.

We now proceed by induction on n . The case $n = 1$ follows immediately from the existence of a zero. For $n > 1$, the factor theorem gives a polynomial Q of degree $n - 1$ such that

$$P(z) = (z - b_1)Q(z).$$

The leading coefficient of Q is a_n . By the induction hypothesis, there are $b_2, \dots, b_n \in \mathbb{C}$ such that

$$Q(z) = a_n \prod_{j=2}^n (z - b_j).$$

Hence

$$P(z) = a_n \prod_{j=1}^n (z - b_j),$$

as required. □

5 Zeros, Convergence, and Maximum Principles

5.1 Zeros of Holomorphic Functions

Definition 5.1

Zero of Order n or ∞

Suppose $f : G \rightarrow \mathbb{C}$ is holomorphic and $f(z_0) = 0$ for some $z_0 \in G$. We say that z_0 is a *zero of finite order* $m \geq 1$ if

$$f(z_0) = f'(z_0) = \dots = f^{(m-1)}(z_0) = 0 \quad \text{but} \quad f^{(m)}(z_0) \neq 0.$$

We say that z_0 is a *zero of order* ∞ if

$$f^{(n)}(z_0) = 0$$

for every $n \geq 0$, where $f^{(0)} = f$.

Thus every zero is either of finite order or of order ∞ , and it can have only one order: the finite order, if it exists, is the least $m \geq 1$ such that $f^{(m)}(z_0) \neq 0$. A zero of order 1 is called a *simple zero*.

Proposition 5.1 Zero of Infinite Order Implies Function is Identically Zero

Let f be holomorphic on a region G , and suppose that

$$f^{(n)}(z_0) = 0$$

for every $n \geq 0$ at some $z_0 \in G$. Then f is identically zero on G .

Proof: Define

$$S = \{z \in G \mid f^{(n)}(z) = 0 \text{ for every } n \geq 0\}.$$

By assumption, $z_0 \in S$, so S is nonempty.

To see that S is open, fix $w \in S$. On some disc $B_r(w) \subseteq G$, the Taylor expansion of f at w gives

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(w)}{n!} (z - w)^n = 0.$$

Thus f vanishes on $B_r(w)$, so every derivative of f also vanishes there. Hence $B_r(w) \subseteq S$.

The set S is closed in G because every derivative $f^{(n)}$ is continuous. Since G is connected and S is nonempty, open, and closed, we have $S = G$. In particular, $f \equiv 0$ on G . $\square$

Proposition 5.2**Zeros of Finite Order Are Isolated**

Let f be holomorphic on an open set $G \subseteq \mathbb{C}$. If $z_0 \in G$ is a zero of f of finite order m , then there exists $r > 0$ such that

$$f(z) \neq 0$$

whenever $0 < |z - z_0| < r$. Thus z_0 is an *isolated zero* of f .

Proof: On a disc centred at z_0 , write the Taylor expansion

$$f(z) = \sum_{n=m}^{\infty} a_n (z - z_0)^n, \quad a_m \neq 0.$$

Factoring out $(z - z_0)^m$ gives

$$f(z) = (z - z_0)^m g(z), \quad g(z) = \sum_{n=m}^{\infty} a_n (z - z_0)^{n-m}.$$

The function g is holomorphic near z_0 and $g(z_0) = a_m \neq 0$. Therefore, by continuity, there is an $r > 0$ such that $g(z) \neq 0$ whenever $|z - z_0| < r$. Hence $f(z) \neq 0$ whenever $0 < |z - z_0| < r$. □

Theorem 5.3**Identity Theorem**

Let G be a region. If $f, g : G \rightarrow \mathbb{C}$ are holomorphic, and if f and g agree on a set with a limit point in G , then $f = g$ on G .

Proof: Let $(z_n)_{n=1}^{\infty}$ be a sequence of distinct points in G such that $z_n \rightarrow z_0 \in G$ and $f(z_n) = g(z_n)$ for every n . Set $h = f - g$. Then h is holomorphic on G , and

$$h(z_n) = 0 \quad \text{for every } n \in \mathbb{N}.$$

By continuity, $h(z_0) = 0$ as well.

Since $z_n \rightarrow z_0$ and the z_n are distinct, every neighbourhood of z_0 contains some zero of h other than z_0 . Thus z_0 is not isolated, so by Proposition 5.2, it is not a zero of finite order.

Since every zero has either finite order or order ∞ , z_0 is a zero of infinite order of h . By Proposition 5.1, we have $h \equiv 0$ on G . Therefore $f = g$ on G . □

Example

Suppose $f : \mathbb{D} \rightarrow \mathbb{C}$ is holomorphic and

$$f\left(\frac{1}{n}\right) = \frac{1}{n^2}$$

for every $n \geq 2$. We can determine $f'(\frac{i}{2})$. Let $g(z) = z^2$. Then $f(\frac{1}{n}) = g(\frac{1}{n})$ for every $n \geq 2$, and $\frac{1}{n} \rightarrow 0 \in \mathbb{D}$. Since the set $\{\frac{1}{n} \mid n \geq 2\}$ has a limit point in $\mathbb{D}$, the Identity Theorem gives $f = g$ on $\mathbb{D}$. Therefore,

$$f'\left(\frac{i}{2}\right) = g'\left(\frac{i}{2}\right) = 2\left(\frac{i}{2}\right) = i.$$

5.2 Weierstrass Convergence Theorem

Theorem 5.4

Weierstrass Convergence Test

Let $G \subseteq \mathbb{C}$ be open, and let $f_k : G \rightarrow \mathbb{C}$ be holomorphic for every $k \in \mathbb{N}$. If $f_k \rightarrow f$ locally uniformly on G , then f is holomorphic on G . Moreover, for every $n \in \mathbb{N}$, $f_k^{(n)} \rightarrow f^{(n)}$ locally uniformly on G .

Example

Weierstrass Convergence Test fails in $\mathbb{R}$

The analogous statement is false for real differentiability. For example, define

$$f_k(x) = \sqrt{x^2 + \frac{1}{k}}, \quad x \in \mathbb{R}.$$

Each f_k is smooth on $\mathbb{R}$, and $f_k \rightarrow |x|$ uniformly on every compact interval. However, $|x|$ is not differentiable at 0.

Proof of Weierstrass Convergence Test: Fix $z_0 \in G$. Since $f_k \rightarrow f$ locally uniformly, there is an open disc $D(z_0, R_0) \subseteq G$ on which the convergence is uniform. Choose $0 < r < R < R_0$, and let C_R be the positively oriented circle $|\zeta - z_0| = R$.

For every $k \in \mathbb{N}$ and every $z \in D(z_0, R)$, Cauchy's integral formula gives

$$f_k(z) = \frac{1}{2\pi i} \int_{C_R} \frac{f_k(\zeta)}{\zeta - z} d\zeta.$$

Since $f_k \rightarrow f$ uniformly on C_R , we may interchange the limit and the path integral. Therefore,

$$f(z) = \frac{1}{2\pi i} \int_{C_R} \frac{f(\zeta)}{\zeta - z} d\zeta$$

for every $z \in D(z_0, R)$. The right-hand side is a Cauchy integral, so it is holomorphic on $D(z_0, R)$. Thus f is holomorphic near z_0 . Since z_0 was arbitrary, f is holomorphic on G .

Now fix $n \in \mathbb{N}$. By Cauchy's formula for derivatives, for $z \in \overline{D}(z_0, r)$,

$$f_k^{(n)}(z) - f^{(n)}(z) = \frac{n!}{2\pi i} \int_{C_R} \frac{f_k(\zeta) - f(\zeta)}{(\zeta - z)^{n+1}} d\zeta.$$

Since $|\zeta - z| \geq R - r$ whenever $\zeta \in C_R$ and $z \in \overline{D}(z_0, r)$, the ML inequality gives

$$\sup_{z \in \overline{D}(z_0, r)} \left| f_k^{(n)}(z) - f^{(n)}(z) \right| \leq \frac{n!R}{(R-r)^{n+1}} \sup_{\zeta \in C_R} |f_k(\zeta) - f(\zeta)|.$$

The right-hand side tends to 0 as $k \rightarrow \infty$. Hence $f_k^{(n)} \rightarrow f^{(n)}$ uniformly on $\overline{D}(z_0, r)$. Since z_0 and n were arbitrary, the convergence of every derivative is locally uniform on G . $\square$

5.3 Maximum Modulus Principle and Schwarz Lemma

Theorem 5.5

Maximum Modulus Principle

If G is a region and $f : G \rightarrow \mathbb{C}$ is holomorphic and non-constant, then $|f|$ has no local maximum in G .

Proof: Suppose, toward a contradiction, that $|f|$ has a local maximum at $z_0 \in G$. Choose $r > 0$ such that

$$\overline{D}(z_0, r) \subseteq G$$

and

$$|f(z)| \leq |f(z_0)|$$

for every $z \in \overline{D}(z_0, r)$.

If $f(z_0) = 0$, then $f = 0$ on $D(z_0, r)$. By the Identity Theorem, f is identically zero on G , contrary to the assumption that f is non-constant. Hence $f(z_0) \neq 0$.

Fix $0 < \rho < r$. By the mean value property,

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + \rho e^{it}) dt.$$

Dividing by $f(z_0)$ gives

$$1 = \frac{1}{2\pi} \int_0^{2\pi} \frac{f(z_0 + \rho e^{it})}{f(z_0)} dt.$$

Taking real parts,

$$1 = \frac{1}{2\pi} \int_0^{2\pi} \operatorname{Re} \left(\frac{f(z_0 + \rho e^{it})}{f(z_0)} \right) dt.$$

But

$$\operatorname{Re} \left(\frac{f(z_0 + \rho e^{it})}{f(z_0)} \right) \leq \left| \frac{f(z_0 + \rho e^{it})}{f(z_0)} \right| \leq 1.$$

The continuous real-valued integrand is everywhere at most 1 and has average 1, so

$$\operatorname{Re} \left(\frac{f(z_0 + \rho e^{it})}{f(z_0)} \right) = 1$$

for every t . Since

$$\left| \frac{f(z_0 + \rho e^{it})}{f(z_0)} \right| \leq 1,$$

the imaginary part must be 0. Hence

$$\frac{f(z_0 + \rho e^{it})}{f(z_0)} = 1$$

for every t , so $f = f(z_0)$ on the circle of radius ρ around z_0 . Since ρ was arbitrary, $f = f(z_0)$ on $D(z_0, r)$. By the Identity Theorem, f is constant on G , a contradiction. $\square$

Exercise 5.3.1

Prove the Fundamental Theorem of Algebra using the Maximum Modulus Principle.

Corollary 5.6

If $f : G \rightarrow \mathbb{C}$ is holomorphic, f is non-constant, and $K \subseteq G$ is compact and non-empty, then the maximum of $|f|$ on K is achieved at some point of ∂K . Equivalently,

$$\max_{z \in K} |f(z)| = \max_{z \in \partial K} |f(z)|.$$

Proof: Since K is compact and $|f|$ is continuous, $|f|$ attains a maximum on K , say at $z_0 \in K$.

If z_0 were in the interior of K , then there would be an $r > 0$ such that $D(z_0, r) \subseteq K$. Hence

$$|f(z)| \leq |f(z_0)|$$

for every $z \in D(z_0, r)$, so $|f|$ would have a local maximum at z_0 . This contradicts the Maximum Modulus Principle, since f is non-constant.

Therefore every point where $|f|$ achieves its maximum value on K lies in $K \setminus \operatorname{int}(K) = \partial K$. Thus the maximum over K is achieved on ∂K . $\square$

Note

Minimum Modulus Principle

A nonconstant holomorphic function can have an interior minimum of its modulus, but only at a zero. For example, if $f(z) = z$ on $\mathbb{D}$, then $|f|$ has its minimum at 0.

Theorem 5.7**Minimum Modulus Principle**

Suppose $f : G \rightarrow \mathbb{C}$ is holomorphic and nonconstant. If $|f|$ has a local minimum at $z_0 \in G$, then $f(z_0) = 0$.

Proof: If $f(z_0) \neq 0$, then f is nonzero on some disk $D(z_0, r) \subseteq G$. Hence $\frac{1}{f}$ is holomorphic on this disk. Since $|f|$ has a local minimum at z_0 ,

$$\left| \frac{1}{f} \right| = \frac{1}{|f|}$$

has a local maximum at z_0 . By the Maximum Modulus Principle, $\frac{1}{f}$ is constant on $D(z_0, r)$, so f is constant on $D(z_0, r)$. The Identity Theorem then forces f to be constant on G , a contradiction. Therefore $f(z_0) = 0$. $\square$

Theorem 5.8**Schwarz's Lemma**

Suppose $f : \mathbb{D} \rightarrow \mathbb{D}$ is holomorphic, where $\mathbb{D}$ is the open unit disc, and suppose $f(0) = 0$. Then

$$|f(z)| \leq |z|$$

for $z \in \mathbb{D}$. Moreover, if $|f(z)| = |z|$ for some $0 \neq z \in \mathbb{D}$, then there exists $\lambda \in \mathbb{C}$, $|\lambda| = 1$ such that $f(z) = \lambda z$ for all $z \in \mathbb{D}$.

Proof: Since f is holomorphic and $f(0) = 0$, local analyticity gives a power series expansion near 0 of the form

$$f(z) = \sum_{n=1}^{\infty} a_n z^n.$$

Define

$$g(z) = \begin{cases} \frac{f(z)}{z}, & \text{if } z \neq 0 \\ f'(0), & \text{if } z = 0. \end{cases}$$

Dividing the power series by z gives, for $z \neq 0$ near 0,

$$\frac{f(z)}{z} = \sum_{n=1}^{\infty} a_n z^{n-1}.$$

The right-hand side is a power series whose value at $z = 0$ is $a_1 = f'(0)$, so it agrees with the definition of g at 0. Thus g is holomorphic at 0. On $\mathbb{D} \setminus \{0\}$, we have $g = \frac{f}{z}$, so g is holomorphic there as well. Hence g is holomorphic on $\mathbb{D}$.

Fix $0 < r < 1$. For $|z| = r$ we have

$$|g(z)| = \frac{|f(z)|}{|z|} \leq \frac{1}{r},$$

because $f(\mathbb{D}) \subseteq \mathbb{D}$. Since g is continuous on the compact disc $|z| \leq r$, $|g|$ attains a maximum there. If the maximum were attained at an interior point and were larger than the boundary maximum, then the Maximum Modulus Principle would force g to be constant. Thus

$$|g(z)| \leq \frac{1}{r}$$

whenever $|z| \leq r$.

Now fix $z \in \mathbb{D}$. Letting $r \rightarrow 1$ with $|z| < r < 1$, we obtain $|g(z)| \leq 1$. Hence, for $z \neq 0$,

$$|f(z)| = |z||g(z)| \leq |z|,$$

and the inequality is also clear at $z = 0$.

Finally, suppose $|f(z_0)| = |z_0|$ for some $z_0 \neq 0$. Then $|g(z_0)| = 1$, so $|g|$ attains its maximum value at an interior point of $\mathbb{D}$. By the Maximum Modulus Principle, g is constant on $\mathbb{D}$; say $g = \lambda$. Since $|g(z_0)| = 1$, we have $|\lambda| = 1$. Therefore $f(z) = \lambda z$ for every $z \in \mathbb{D}$. □

6 Laurent Series and Singularities

6.1 Laurent Series

Note

Notation

For $z_0 \in \mathbb{C}$ and $r > 0$, write

$$D(z_0, r) := \{z \in \mathbb{C} \mid |z - z_0| < r\}$$

for the disk centered at z_0 with radius r , and write

$$D^*(z_0, r) := \{z \in \mathbb{C} \mid 0 < |z - z_0| < r\}$$

for the corresponding punctured disk.

For $r > 0$, write

$$C_r(z_0) := \{z \in \mathbb{C} \mid |z - z_0| = r\}$$

for the circle centered at z_0 with radius r . When $C_r(z_0)$ is used as a contour, it is oriented counter-clockwise unless stated otherwise.

For $0 \leq r_1 < r_2 \leq \infty$, write

$$A(z_0, r_1, r_2) := \{z \in \mathbb{C} \mid r_1 < |z - z_0| < r_2\}$$

for the annulus centered at z_0 with inner radius r_1 and outer radius r_2 . Thus $D^*(z_0, r) = A(z_0, 0, r)$.

Example

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n$$

for $|z| < 1$. Moreover,

$$\begin{aligned} \frac{1}{1-z} &= -\frac{1}{z} \left(\frac{1}{1-\frac{1}{z}} \right) \\ &= -\frac{1}{z} \sum_{n=0}^{\infty} \left(\frac{1}{z} \right)^n \\ &= \sum_{n=-1}^{-\infty} -z^n \end{aligned}$$

for $|z| > 1$.

Example

$$\frac{1}{(z-1)(z-2)} = \frac{1}{z-2} - \frac{1}{z-1}$$

Note that

$$\begin{aligned}\frac{1}{z-2} &= -\frac{1}{2} \left(\frac{1}{1 - \frac{z}{2}} \right) \\ &= -\frac{1}{2} \sum_{n=0}^{\infty} 2^{-n} z^n\end{aligned}$$

for $|z| < 2$. Also,

$$-\frac{1}{z-1} = \sum_{n=-1}^{-\infty} -z^n$$

for $|z| > 1$. Therefore,

$$\frac{1}{(z-1)(z-2)} = \sum_{n=-1}^{-\infty} -z^n - \frac{1}{2} \sum_{n=0}^{\infty} 2^{-n} z^n$$

for $1 < |z| < 2$.

Definition 6.1**Laurent Series**

Let $z_0 \in \mathbb{C}$ and let $(a_n)_{n \in \mathbb{Z}}$ be a bi-infinite sequence in $\mathbb{C}$ (equivalently, a function $\mathbb{Z} \rightarrow \mathbb{C}$). A *Laurent series* centered at z_0 with coefficients $(a_n)_{n \in \mathbb{Z}}$ is a formal power series of the form

$$\sum_{n=-\infty}^{\infty} a_n (z - z_0)^n.$$

We say that the Laurent series converges at z if both

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n \quad \text{and} \quad \sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n}$$

converge. In that case,

$$\sum_{n=-\infty}^{\infty} a_n (z - z_0)^n := \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n}.$$

Note**Symmetric Partial Sums**

The *symmetric partial sums* of a Laurent series are

$$S_N(z) := \sum_{n=-N}^N a_n(z - z_0)^n.$$

If the Laurent series converges at z , then the symmetric partial sums converge, and

$$\lim_{N \rightarrow \infty} S_N(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n + \sum_{n=1}^{\infty} a_{-n}(z - z_0)^{-n}.$$

Theorem 6.1**Annulus of Convergence for Laurent Series**

Let $z_0 \in \mathbb{C}$, and consider the Laurent series

$$\sum_{n=-\infty}^{\infty} a_n(z - z_0)^n.$$

Define

$$R_2 := \frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}}$$

$$R_1 := \limsup_{n \rightarrow \infty} |a_{-n}|^{\frac{1}{n}}$$

with the convention that $\frac{1}{0} = \infty$ and $\frac{1}{\infty} = 0$ in the definition of R_2 . Then

1. The Laurent series converges absolutely and locally uniformly on the annulus

$$A(z_0, R_1, R_2).$$

2. The function

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$$

is holomorphic on $A(z_0, R_1, R_2)$.

Proof: The non-negative part

$$\sum_{n=0}^{\infty} a_n(z - z_0)^n$$

is an ordinary power series with radius of convergence R_2 , by the Cauchy-Hadamard theorem.

For the negative part, put $w = \frac{1}{z - z_0}$. Then

$$\sum_{n=-1}^{-\infty} a_n (z - z_0)^n = \sum_{m=1}^{\infty} a_{-m} w^m.$$

This is a power series in w with radius of convergence $\frac{1}{R_1}$. Hence it converges whenever

$$|w| < \frac{1}{R_1},$$

which is equivalent to $|z - z_0| > R_1$.

Therefore both parts converge whenever $z \in A(z_0, R_1, R_2)$.

We now check local uniform convergence. Let $K \subseteq A(z_0, R_1, R_2)$ be compact. Since $z \mapsto |z - z_0|$ is continuous, the image

$$\{|z - z_0| : z \in K\}$$

is a compact subset of (R_1, R_2) . Hence there exist numbers ρ_1, ρ_2 such that

$$R_1 < \rho_1 \leq |z - z_0| \leq \rho_2 < R_2$$

for every $z \in K$.

The positive part is a power series with radius of convergence R_2 , so it converges uniformly on $|z - z_0| \leq \rho_2$. Hence it converges uniformly on K . For the negative part, the set K is sent by $w = \frac{1}{z - z_0}$ into

$$|w| \leq \frac{1}{\rho_1} < \frac{1}{R_1}.$$

Since the negative part is a power series in w with radius of convergence $\frac{1}{R_1}$, it converges uniformly on $|w| \leq \frac{1}{\rho_1}$, and therefore uniformly on K .

Thus both parts converge uniformly on every compact subset of $A(z_0, R_1, R_2)$, so the Laurent series converges locally uniformly there. Since each symmetric partial sum is holomorphic on $A(z_0, R_1, R_2)$, the Weierstrass Convergence Theorem implies that the sum is holomorphic there. □

Theorem 6.2**Differentiating Laurent Series**

Suppose

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$$

on the annulus $A(z_0, R_1, R_2)$. Then f may be differentiated term by term on this annulus, and

$$f'(z) = \sum_{n=-\infty}^{\infty} n a_n(z - z_0)^{n-1}.$$

Moreover, the differentiated Laurent series has the same annulus of convergence.

Proof: The non-negative part is an ordinary power series, so it may be differentiated term by term and its radius of convergence remains R_2 .

For the negative part, write $w = \frac{1}{z - z_0}$. Then

$$\sum_{n=-\infty}^{-1} a_n(z - z_0)^n = \sum_{m=1}^{\infty} a_{-m} w^m.$$

This is a power series in w , so it may be differentiated term by term in w and its radius of convergence remains $\frac{1}{R_1}$. Since $w = (z - z_0)^{-1}$ is holomorphic on the annulus, the chain rule gives

$$\frac{d}{dz}(a_{-m}(z - z_0)^{-m}) = -m a_{-m}(z - z_0)^{-m-1}.$$

Combining this with the differentiated non-negative part gives

$$f'(z) = \sum_{n=-\infty}^{\infty} n a_n(z - z_0)^{n-1}.$$

The convergence radii of the two differentiated pieces are unchanged, so the annulus of convergence is again $A(z_0, R_1, R_2)$. □

Theorem 6.3**Cauchy's Theorem for the Annulus**

Let $0 \leq R_1 < R_2 \leq \infty$, and let f be holomorphic on $A(z_0, R_1, R_2)$. If $R_1 < r < s < R_2$, then

$$\int_{C_r(z_0)} f(z) dz = \int_{C_s(z_0)} f(z) dz.$$

Equivalently, the integral of f around the boundary of the closed annulus $\{z \in \mathbb{C} \mid r \leq |z - z_0| \leq s\}$ is zero, with the boundary orientation.

Proof: It suffices to prove the equivalent boundary statement. Fix $N \geq 3$ and set

$$\theta_k = 2\pi \frac{k}{N}, \quad p_k = z_0 + se^{i\theta_k}, \quad q_k = z_0 + re^{i\theta_k}$$

for $k = 0, \dots, N$, where $p_N = p_0$ and $q_N = q_0$. Let P_s^N be the polygonal path through $p_0, p_1, \dots, p_N$, oriented counter-clockwise, and let P_r^N be the polygonal path through $q_0, q_1, \dots, q_N$, also oriented counter-clockwise.

For each $k = 0, \dots, N-1$, let T_k be the closed quadrilateral with vertices

$$q_k, p_k, p_{k+1}, q_{k+1}.$$

This quadrilateral lies in the annulus, and splitting it along a diagonal expresses it as the union of two triangles. Since f is holomorphic on a neighbourhood of each triangle, Cauchy's theorem for triangles gives

$$\int_{\partial T_k} f(z) dz = 0$$

with positive boundary orientation.

Summing over k , the integrals over the radial cuts cancel in pairs, because each cut is traversed once in each direction. The remaining outer boundary is P_s^N , while the remaining inner boundary is $-P_r^N$. Therefore

$$\int_{P_s^N} f(z) dz - \int_{P_r^N} f(z) dz = 0,$$

so

$$\int_{P_s^N} f(z) dz = \int_{P_r^N} f(z) dz.$$

As $N \rightarrow \infty$, the polygonal paths P_s^N and P_r^N converge uniformly to the corresponding circles $C_s(z_0)$ and $C_r(z_0)$, and their lengths remain bounded. Since f is uniformly continuous on the compact annulus $\{z \in \mathbb{C} \mid r \leq |z - z_0| \leq s\}$, the polygonal integrals converge to the circular integrals. Passing to the limit gives

$$\int_{C_s(z_0)} f(z) dz = \int_{C_r(z_0)} f(z) dz,$$

as required. □

Proof: Alternative proof. For $R_1 < r < R_2$, let

$$I(r) = \int_{C_r(z_0)} f(z) dz.$$

Using the parametrization

$$t \mapsto z_0 + re^{it}, \quad 0 \leq t \leq 2\pi,$$

of $C_r(z_0)$, we can write

$$I(r) = \int_0^{2\pi} f(z_0 + re^{it}) ire^{it} dt.$$

The integrand in the preceding integral, as a function of the two real variables r and t , has continuous partial derivatives. Thus we may differentiate under the integral sign:

$$\frac{dI(r)}{dr} = \int_0^{2\pi} \partial_r [f(z_0 + re^{it}) ire^{it}] dt.$$

We have

$$\begin{aligned} \partial_r [f(z_0 + re^{it}) ire^{it}] &= f'(z_0 + re^{it}) ire^{2it} + f(z_0 + re^{it}) ie^{it} \\ &= \partial_t [f(z_0 + re^{it}) e^{it}]. \end{aligned}$$

Consequently,

$$\begin{aligned} \frac{dI(r)}{dr} &= \int_0^{2\pi} \partial_t [f(z_0 + re^{it}) e^{it}] dt \\ &= [f(z_0 + re^{it}) e^{it}]_{t=0}^{t=2\pi} \\ &= 0. \end{aligned}$$

Hence I is constant on (R_1, R_2) , and therefore

$$\int_{C_r(z_0)} f(z) dz = \int_{C_s(z_0)} f(z) dz$$

whenever $R_1 < r < s < R_2$. □

6.2 Laurent Series Representations

Theorem 6.4

Cauchy's Formula for an Annulus

Let $0 \leq R_1 < R_2 \leq \infty$, let $z_0 \in \mathbb{C}$, and put $\Omega = A(z_0, R_1, R_2)$. Suppose that f is holomorphic on Ω . For $R_1 < r < R_2$, let $C_r(z_0)$ be as in the notation above. If $w \in \Omega$ and

$$R_1 < r_1 < |w - z_0| < r_2 < R_2,$$

then

$$f(w) = \frac{1}{2\pi i} \int_{C_{r_2}(z_0)} \frac{f(z)}{z - w} dz - \frac{1}{2\pi i} \int_{C_{r_1}(z_0)} \frac{f(z)}{z - w} dz.$$

Proof: Fix w, r_1 , and r_2 as in the statement. Define $g : \Omega \rightarrow \mathbb{C}$ by

$$g(z) = \begin{cases} \frac{f(z)-f(w)}{z-w}, & z \neq w \\ f'(w), & z = w. \end{cases}$$

Then g is holomorphic on Ω : it is holomorphic on $\Omega \setminus \{w\}$, and the value $g(w) = f'(w)$ removes the apparent singularity at w .

By Cauchy's theorem for the annulus applied to g on Ω ,

$$\int_{C_{r_1}(z_0)} g(z) dz = \int_{C_{r_2}(z_0)} g(z) dz.$$

Expanding g and moving terms gives

$$\int_{C_{r_2}(z_0)} \frac{f(z)}{z-w} dz - \int_{C_{r_1}(z_0)} \frac{f(z)}{z-w} dz = f(w) \int_{C_{r_2}(z_0)} \frac{1}{z-w} dz - f(w) \int_{C_{r_1}(z_0)} \frac{1}{z-w} dz.$$

Since $|w - z_0| < r_2$, the point w lies inside $C_{r_2}(z_0)$. Hence Cauchy's formula for the circle, applied to the constant function 1, gives

$$\int_{C_{r_2}(z_0)} \frac{1}{z-w} dz = 2\pi i.$$

Since $|w - z_0| > r_1$, the function $z \mapsto \frac{1}{z-w}$ is holomorphic on a neighbourhood of the closed disk $\{z \in \mathbb{C} \mid |z - z_0| \leq r_1\}$. By Cauchy's theorem for convex regions,

$$\int_{C_{r_1}(z_0)} \frac{1}{z-w} dz = 0.$$

Therefore

$$\int_{C_{r_2}(z_0)} \frac{f(z)}{z-w} dz - \int_{C_{r_1}(z_0)} \frac{f(z)}{z-w} dz = 2\pi i f(w),$$

and division by $2\pi i$ gives the formula. □

Theorem 6.5

Existence of Laurent Series Representations

Let $0 \leq R_1 < R_2 \leq \infty$, let $z_0 \in \mathbb{C}$, and put $\Omega = A(z_0, R_1, R_2)$. If f is holomorphic on Ω , then f has a Laurent series representation on Ω :

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n, \quad z \in \Omega,$$

where, for any r satisfying $R_1 < r < R_2$,

$$a_n = \frac{1}{2\pi i} \int_{C_r(z_0)} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta, \quad n \in \mathbb{Z}.$$

Proof: First note that the coefficient formula is independent of the admissible radius r . Indeed, for fixed $n \in \mathbb{Z}$, the function

$$\zeta \mapsto \frac{f(\zeta)}{(\zeta - z_0)^{n+1}}$$

is holomorphic on Ω . Hence Cauchy's theorem for the annulus shows that its integrals over any two centered circles lying in Ω are equal.

Now fix $z \in \Omega$. Choose radii r_1, r_2 such that

$$R_1 < r_1 < |z - z_0| < r_2 < R_2.$$

By Cauchy's formula for an annulus,

$$f(z) = \frac{1}{2\pi i} \int_{C_{r_2}(z_0)} \frac{f(\zeta)}{\zeta - z} d\zeta - \frac{1}{2\pi i} \int_{C_{r_1}(z_0)} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

On $C_{r_2}(z_0)$ we have $|z - z_0| < |\zeta - z_0| = r_2$, so

$$\begin{aligned} \frac{1}{\zeta - z} &= \frac{1}{\zeta - z_0} \cdot \frac{1}{1 - \frac{z - z_0}{\zeta - z_0}} \\ &= \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\zeta - z_0)^{n+1}}, \end{aligned}$$

uniformly for $\zeta \in C_{r_2}(z_0)$. Therefore

$$\frac{1}{2\pi i} \int_{C_{r_2}(z_0)} \frac{f(\zeta)}{\zeta - z} d\zeta = \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \int_{C_{r_2}(z_0)} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta \right) (z - z_0)^n.$$

On $C_{r_1}(z_0)$ we have $|\zeta - z_0| = r_1 < |z - z_0|$, so

$$\begin{aligned} \frac{1}{\zeta - z} &= -\frac{1}{z - z_0} \cdot \frac{1}{1 - \frac{\zeta - z_0}{z - z_0}} \\ &= -\sum_{m=0}^{\infty} \frac{(\zeta - z_0)^m}{(z - z_0)^{m+1}}, \end{aligned}$$

uniformly for $\zeta \in C_{r_1}(z_0)$. Hence

$$-\frac{1}{2\pi i} \int_{C_{r_1}(z_0)} \frac{f(\zeta)}{\zeta - z} d\zeta = \sum_{m=0}^{\infty} \left(\frac{1}{2\pi i} \int_{C_{r_1}(z_0)} f(\zeta) (\zeta - z_0)^m d\zeta \right) (z - z_0)^{-m-1}.$$

Re-indexing the last sum by $n = -m - 1$ gives precisely the negative-power part of the Laurent series, because

$$(\zeta - z_0)^m = (\zeta - z_0)^{-n-1}.$$

Combining the two expansions gives

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n.$$

Since the coefficients are independent of the chosen radii r_1 and r_2 , this representation holds for every $z \in \Omega$. The same argument, with r_1, r_2 chosen uniformly around a compact subannulus, also gives locally uniform convergence on Ω . $\square$

6.3 Isolated Singularities

Definition 6.2

Suppose $f : G \rightarrow \mathbb{C}$ is a holomorphic function. A point z_0 is an *isolated singularity* if $z_0 \notin G$, but there exists $R > 0$ such that $D^*(z_0, R) \subseteq G$.

Note

Isolated Singularity

If z_0 is an isolated singularity, then f is holomorphic on some punctured disk $D^*(z_0, R) = A(z_0, 0, R)$. Since this is an annulus, the Laurent series theorem gives a Laurent series representation for f there.

Example

$$f(z) = z \text{ on } G = \{z \in \mathbb{C} \mid z \neq 0\}.$$

Definition 6.3

Removable Singularity

Let $f : G \rightarrow \mathbb{C}$ be holomorphic. We say that $z_0 \notin G$ is a removable singularity if there exists a holomorphic extension of f on $G \cup \{z_0\}$.

Example

Let $f(z) = \frac{z}{e^z - 1}$ on $G = \{z \in \mathbb{C} \mid z \neq 0\}$. Since

$$e^z - 1 = \sum_{n=1}^{\infty} \frac{1}{n!} z^n,$$

we have, for $z \neq 0$,

$$f(z) = \frac{1}{\sum_{n=1}^{\infty} \frac{1}{n!} z^{n-1}}.$$

The denominator is holomorphic near 0 and has value 1 at 0, so this formula defines a holomorphic extension across 0. Therefore 0 is a removable singularity.

Definition 6.4

An isolated singularity is a *pole of order m* if in the Laurent series representation near z_0 , we have

$$f(z) = \sum_{n=-m}^{\infty} a_n(z - z_0)^n$$

with $a_{-m} \neq 0$.

Order 1 poles are *simple poles*.

Example

$f(z) = \frac{1}{z}$ is a simple pole.

Proposition 6.6**Characterization of Poles**

Let $f : G \rightarrow \mathbb{C}$ be holomorphic, and let z_0 be an isolated singularity of f . Then z_0 is a pole of order m if and only if there is a holomorphic function g near z_0 such that

$$f(z) = (z - z_0)^{-m} g(z)$$

for $z \neq z_0$ near z_0 , with $g(z_0) \neq 0$.

Proof: Suppose z_0 is a pole of order m . Then the Laurent series of f near z_0 has the form

$$f(z) = \sum_{n=-m}^{\infty} a_n(z - z_0)^n,$$

with $a_{-m} \neq 0$. Hence

$$f(z) = (z - z_0)^{-m} \sum_{n=0}^{\infty} a_{n-m}(z - z_0)^n$$

The power series

$$g(z) = \sum_{n=0}^{\infty} a_{n-m}(z - z_0)^n$$

is holomorphic near z_0 , and $g(z_0) = a_{-m} \neq 0$.

Conversely, if

$$f(z) = (z - z_0)^{-m} g(z)$$

with g holomorphic near z_0 and $g(z_0) \neq 0$, then the Taylor series of g at z_0 has nonzero constant term. Multiplying by $(z - z_0)^{-m}$ gives a Laurent series whose first nonzero negative-power term is $(z - z_0)^{-m}$. Thus z_0 is a pole of order m . $\square$

Definition 6.5**Essential Singularity**

An isolated singularity z_0 is an *essential singularity* if the Laurent series of f near z_0 has infinitely many nonzero negative-power terms.

Example

The function $f(z) = e^{\frac{1}{z}}$ has an essential singularity at 0, since

$$e^{\frac{1}{z}} = \sum_{n=0}^{\infty} \frac{1}{n!} z^{-n}$$

has infinitely many nonzero negative-power terms.

Definition 6.6 **∞ as an Isolated Singularity**

Let $f : G \rightarrow \mathbb{C}$ be holomorphic. We say that ∞ is an isolated singularity for f if there exists $R > 0$ such that

$$\{z \in \mathbb{C} \mid |z| > R\} \subseteq G.$$

Definition 6.7**Type of Singularity at ∞**

Suppose ∞ is an isolated singularity for f . Define

$$g(w) = f\left(\frac{1}{w}\right)$$

for $w \in D^*\left(0, \frac{1}{R}\right)$. We say that ∞ is a removable singularity, a pole of order m , or an essential singularity for f if 0 has the corresponding type of isolated singularity for g .

Example

The function $f(z) = z$ has a simple pole at ∞ , since

$$f\left(\frac{1}{w}\right) = \frac{1}{w}$$

has a simple pole at 0.

Example

The function $f(z) = \frac{1}{z}$ has a removable singularity at ∞ , since

$$f\left(\frac{1}{w}\right) = w$$

has a removable singularity at 0.

6.4 Removable Singularities

Note

We fix a holomorphic function $f : G \rightarrow \mathbb{C}$, where G is an open set.

Theorem 6.7

Removable Singularity Theorem

Let z_0 be an isolated singularity of $f : G \rightarrow \mathbb{C}$. Then z_0 is removable if and only if f is bounded on some punctured disk around z_0 . That is, there exist $r > 0$ and $M > 0$ such that $D^*(z_0, r) \subseteq G$ and $|f(z)| \leq M$ whenever $z \in D^*(z_0, r)$.

Proof: First suppose z_0 is removable. Let $F : G \cup \{z_0\} \rightarrow \mathbb{C}$ be a holomorphic extension of f . Since F is continuous at z_0 , there is some $r > 0$ such that F is bounded on compact set $\overline{D}(z_0, r)$. Hence f is bounded on $D^*(z_0, r)$.

Conversely, suppose $|f(z)| \leq M$ whenever $z \in D^*(z_0, r)$. Write the Laurent series of f near z_0 as

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n.$$

We show that all negative coefficients vanish. Fix $n \geq 1$. For $0 < \rho < r$, by the Laurent coefficient formula applied on $C_\rho(z_0)$,

$$a_{-n} = \frac{1}{2\pi i} \int_{C_\rho(z_0)} f(z) (z - z_0)^{n-1} dz.$$

Therefore, by the ML inequality,

$$|a_{-n}| \leq \frac{1}{2\pi} \cdot M \rho^{n-1} \cdot 2\pi \rho = M \rho^n.$$

Since this holds for every $0 < \rho < r$, letting $\rho \rightarrow 0$ gives $a_{-n} = 0$. Thus $a_k = 0$ for all $k < 0$, so by the removable singularity criterion, z_0 is removable. $\square$

Theorem 6.8

Pole Characterization by Growth

Suppose z_0 is an isolated singularity of f . Then z_0 is a pole if and only if

$$\lim_{z \rightarrow z_0} |f(z)| = \infty.$$

Proof: First suppose z_0 is a pole of order m . By the characterization of poles, the function

$$g(z) = (z - z_0)^m f(z)$$

has a removable singularity at z_0 , and its removable extension satisfies $g(z_0) \neq 0$. Hence there are $r > 0$ and $c > 0$ such that $|g(z)| \geq c$ whenever $z \in D(z_0, r)$. Thus, for $z \in D^*(z_0, r)$,

$$|f(z)| = \frac{|g(z)|}{|z - z_0|^m} \geq \frac{c}{|z - z_0|^m}.$$

Letting $z \rightarrow z_0$ gives $|f(z)| \rightarrow \infty$.

Conversely, suppose

$$\lim_{z \rightarrow z_0} |f(z)| = \infty.$$

Then there is $r > 0$ such that $f(z) \neq 0$ whenever $z \in D^*(z_0, r)$. Define $h(z) = \frac{1}{f(z)}$ on $D^*(z_0, r)$. Since $|h(z)| \rightarrow 0$ as $z \rightarrow z_0$, the function h is bounded near z_0 , so by the removable singularity theorem, h has a removable singularity at z_0 . Define its extension by $h(z_0) = 0$.

The zero of h at z_0 has finite order. Indeed, h is not identically zero on any punctured disk around z_0 , since f is finite there, so h cannot have a zero of infinite order at z_0 . Thus

$$h(z) = (z - z_0)^m k(z)$$

for some $m \geq 1$, where k is holomorphic near z_0 and $k(z_0) \neq 0$. Therefore

$$f(z) = \frac{1}{h(z)} = (z - z_0)^{-m} \frac{1}{k(z)},$$

where $\frac{1}{k}$ is holomorphic near z_0 . By the characterization of poles, z_0 is a pole of order m . $\square$

Theorem 6.9

Great Picard's Theorem

Let z_0 be an essential singularity of $f : G \rightarrow \mathbb{C}$. For every $\varepsilon > 0$ such that $D^*(z_0, \varepsilon) \subseteq G$, the image $f(D^*(z_0, \varepsilon))$ omits at most one complex number.

Note

The proof of Picard's theorem is out of scope.

Theorem 6.10

Casorati-Weierstrass Theorem

Let z_0 be an essential singularity of $f : G \rightarrow \mathbb{C}$. For every $\varepsilon > 0$ such that $D^*(z_0, \varepsilon) \subseteq G$, the image $f(D^*(z_0, \varepsilon))$ is dense in $\mathbb{C}$.

Proof: Suppose not. Then there exist $\varepsilon > 0$, $w \in \mathbb{C}$, and $\delta > 0$ such that

$$|f(z) - w| \geq \delta$$

whenever $z \in D^*(z_0, \varepsilon)$. In particular, $f(z) \neq w$ on this punctured disk, so

$$g(z) = \frac{1}{f(z) - w}$$

is holomorphic there. Moreover, $|g(z)| \leq \frac{1}{\delta}$, so by the removable singularity theorem, g has a removable singularity at z_0 .

Let G_0 be the holomorphic extension of g to a disk centered at z_0 . We have two cases:

1. If $G_0(z_0) \neq 0$, then $\frac{1}{G_0}$ is holomorphic in a smaller disk centered at z_0 , so

$$f(z) = w + \frac{1}{G_0}(z)$$

has a removable singularity at z_0 , a contradiction.

2. If $G_0(z_0) = 0$, then $|g(z)| \rightarrow 0$ as $z \rightarrow z_0$, so

$$|f(z) - w| = \frac{1}{|g(z)|} \rightarrow \infty.$$

By the pole characterization by growth, $f - w$ has a pole at z_0 , and hence f has a pole at z_0 , again a contradiction.

Therefore the image of every punctured disk around z_0 is dense in $\mathbb{C}$. □

7 Global Cauchy Theory and Residues

Note

Goal

We want to extend Cauchy's theorem from convex regions to simply connected regions. Convexity was useful because every triangle and line segment stayed inside the region. For simply connected regions, we need a topological replacement for that condition. The next results set this up using continuous logarithms along curves, increments of argument, and winding numbers.

7.1 Continuous Logarithms Along Curves

Lemma 7.1

Partition Lemma

Let $a < b$, and let $\varphi : [a, b] \rightarrow \mathbb{C}^*$ be continuous. Then there are a partition

$$a = t_0 < t_1 < \dots < t_n = b$$

and open disks $D_0, \dots, D_{n-1}$, none of which contains 0, such that

$$\varphi([t_j, t_{j+1}]) \subseteq D_j$$

for every $j = 0, \dots, n-1$.

Proof: Since φ is continuous on the compact interval $[a, b]$, it is uniformly continuous. Moreover, $\varphi(t) \neq 0$ for every $t \in [a, b]$, so the Extreme Value Theorem gives

$$m := \min_{t \in [a, b]} |\varphi(t)| > 0.$$

By uniform continuity, there exists $\delta > 0$ such that

$$|s - t| < \delta \implies |\varphi(s) - \varphi(t)| < \frac{m}{2}.$$

Choose a partition whose mesh is smaller than δ , and set

$$D_j = D\left(\varphi(t_j), \frac{m}{2}\right).$$

If $t \in [t_j, t_{j+1}]$, then

$$|\varphi(t) - \varphi(t_j)| < \frac{m}{2},$$

so $\varphi([t_j, t_{j+1}]) \subseteq D_j$. Finally, $|\varphi(t_j)| \geq m > \frac{m}{2}$, so $0 \notin D_j$. $\square$

Theorem 7.2

Path Lifting of The Exponential

Let $a < b$, and let $\varphi : [a, b] \rightarrow \mathbb{C}^*$ be continuous. Then there exists a continuous map $\psi : [a, b] \rightarrow \mathbb{C}$ such that

$$e^{\psi(t)} = \varphi(t)$$

for every $t \in [a, b]$. Moreover, ψ is unique up to adding a constant in $2\pi i\mathbb{Z}$.

Proof: By the Partition Lemma, choose a partition

$$a = t_0 < t_1 < \dots < t_n = b$$

and disks $D_0, \dots, D_{n-1}$ such that $\varphi([t_j, t_{j+1}]) \subseteq D_j$ and $0 \notin D_j$ for every j . For each j , choose a branch L_j of $\log$ on D_j , and define

$$\psi_j = L_j \circ \varphi$$

on $[t_j, t_{j+1}]$. Then ψ_j is continuous and

$$e^{\psi_j(t)} = \varphi(t).$$

We now adjust these local lifts inductively. Set $\tilde{\psi}_0 = \psi_0$. Suppose $\tilde{\psi}_{j-1}$ has been defined on $[t_{j-1}, t_j]$. At the common endpoint t_j ,

$$e^{\tilde{\psi}_{j-1}(t_j)} = \varphi(t_j) = e^{\psi_j(t_j)}.$$

Therefore

$$c_j := \tilde{\psi}_{j-1}(t_j) - \psi_j(t_j) \in 2\pi i\mathbb{Z}.$$

Define

$$\tilde{\psi}_j(t) := \psi_j(t) + c_j$$

for $t \in [t_j, t_{j+1}]$. Then

$$\tilde{\psi}_j(t_j) = \tilde{\psi}_{j-1}(t_j)$$

and, since $e^{c_j} = 1$,

$$e^{\tilde{\psi}_j(t)} = e^{\psi_j(t)} = \varphi(t).$$

Continuing for $j = 1, \dots, n-1$ gives local lifts that agree at every partition point. Define ψ by setting $\psi(t) = \tilde{\psi}_j(t)$ on each $[t_j, t_{j+1}]$. The finite pasting lemma shows that ψ is continuous on $[a, b]$, and the preceding identity gives $e^{\psi(t)} = \varphi(t)$ everywhere.

If ψ and η are two such choices, then $\psi - \eta$ is a continuous map from $[a, b]$ to the discrete set $2\pi i\mathbb{Z}$. Since $[a, b]$ is connected, $\psi - \eta$ is constant. $\square$

Example

Piecewise- C^1 Formula

Let $\varphi : [a, b] \rightarrow \mathbb{C}^*$ be piecewise- C^1 . Fix $c \in \mathbb{C}$ such that $e^c = \varphi(a)$, and set

$$\psi(t) = c + \int_a^t \frac{\varphi'(s)}{\varphi(s)} ds.$$

Then ψ is continuous, piecewise- C^1 , and $\psi'(t) = \frac{\varphi'(t)}{\varphi(t)}$ wherever φ' exists. Hence $(\varphi e^{-\psi})' = 0$ on each smooth piece, and since $\varphi(a)e^{-\psi(a)} = 1$, we get

$$e^{\psi(t)} = \varphi(t)$$

for every $t \in [a, b]$. Thus ψ is a continuous logarithm of φ , and its increment is

$$\psi(b) - \psi(a) = \int_a^b \frac{\varphi'(t)}{\varphi(t)} dt.$$

Corollary 7.3

Continuous Logarithm of a Pullback Along a Curve

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be continuous, and let f be continuous on $R(\gamma)$. If

$$(f \circ \gamma)(t) \neq 0$$

for every $t \in [a, b]$, then there exists a continuous map $\psi : [a, b] \rightarrow \mathbb{C}$ such that

$$e^{\psi(t)} = (f \circ \gamma)(t)$$

for every $t \in [a, b]$.

Proof: Let $\varphi = f \circ \gamma$. Since γ is continuous and f is continuous on $R(\gamma)$, the map $\varphi : [a, b] \rightarrow \mathbb{C}$ is continuous. By assumption,

$$\varphi(t) = (f \circ \gamma)(t) \neq 0$$

for every $t \in [a, b]$, so φ is a continuous map from $[a, b]$ into $\mathbb{C}^*$.

By Theorem 7.2, there exists a continuous map $\psi : [a, b] \rightarrow \mathbb{C}$ such that $e^{\psi(t)} = \varphi(t)$ for every $t \in [a, b]$. Since $\varphi = f \circ \gamma$, this gives

$$e^{\psi(t)} = (f \circ \gamma)(t)$$

for every $t \in [a, b]$. □

7.2 Increments and Winding Numbers

Definition 7.1

Increment in Log

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a continuous curve, and let f be continuous and nonzero on $R(\gamma)$. Choose a continuous map $\psi : [a, b] \rightarrow \mathbb{C}$ such that

$$e^{\psi(t)} = (f \circ \gamma)(t)$$

for every $t \in [a, b]$. The *increment in log* of f along γ is

$$\Delta(\log f, \gamma) = \psi(b) - \psi(a).$$

Note

Increment in Log is well-defined: if $\tilde{\psi} : [a, b] \rightarrow \mathbb{C}$ is another continuous map with $e^{\tilde{\psi}(t)} = (f \circ \gamma)(t)$ for every $t \in [a, b]$, then there exists $m \in \mathbb{Z}$ such that

$$\tilde{\psi}(t) = \psi(t) + 2\pi im$$

for every $t \in [a, b]$. Hence

$$\tilde{\psi}(b) - \tilde{\psi}(a) = (\psi(b) + 2\pi im) - (\psi(a) + 2\pi im) = \psi(b) - \psi(a).$$

Example

Let $\gamma(t) = e^{it}$ for $0 \leq t \leq 2\pi$, and let $f(z) = z$. We can take $\psi(t) = it$, so

$$\Delta(\log z, \gamma) = \psi(2\pi) - \psi(0) = 2\pi i.$$

Definition 7.2

Increment in Argument

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a continuous curve, and let f be continuous and nonzero on $R(\gamma)$. The *increment in argument* of f along γ is

$$\Delta(\arg f, \gamma) := \operatorname{Im}(\Delta(\log f, \gamma)) \in \mathbb{R}.$$

Definition 7.3**Winding Number**

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a continuous closed curve, and let $z_0 \notin R(\gamma)$. The *winding number* of γ about z_0 is

$$\text{Ind}_{z=z_0}(\gamma) = \frac{1}{2\pi} \Delta(\arg(z - z_0), \gamma).$$

This is an integer.

Example

Let $\gamma(t) = e^{it}$ for $0 \leq t \leq 2\pi$. For $z_0 = 0$, we have

$$\Delta(\arg(z - 0), \gamma) = \Delta(\arg z, \gamma) = 2\pi.$$

Hence

$$\text{Ind}_{z=0}(\gamma) = \frac{1}{2\pi} \cdot 2\pi = 1.$$

Note

If γ is closed and f is continuous and nonzero on $R(\gamma)$, then $(f \circ \gamma)(a) = (f \circ \gamma)(b)$. Thus the starting and ending values have the same magnitude, so the real part of $\Delta(\log f, \gamma)$ is 0. The only possible net change is in the continuously chosen argument, which can change by an integer multiple of 2π . Hence $\Delta(\log f, \gamma)$ is an integer multiple of $2\pi i$, and

$$\Delta(\arg f, \gamma) = \frac{1}{i} \Delta(\log f, \gamma)$$

is an integer multiple of 2π .

Example

Let $\gamma : [a, b] \rightarrow \mathbb{C}$ be a piecewise- C^1 closed curve, and let $z_0 \notin R(\gamma)$. Set $\varphi(t) = \gamma(t) - z_0$. By the piecewise- C^1 formula for a continuous logarithm along a curve, we may choose ψ so that $e^{\psi(t)} = \varphi(t)$ and

$$\psi(b) - \psi(a) = \int_a^b \frac{\varphi'(t)}{\varphi(t)} dt = \int_a^b \frac{\gamma'(t)}{\gamma(t) - z_0} dt.$$

Therefore $\Delta(\log(z - z_0), \gamma) = \int_a^b \frac{\gamma'(t)}{\gamma(t) - z_0} dt$. Therefore

$$\text{Ind}_{z=z_0}(\gamma) = \frac{1}{2\pi i} \int_a^b \frac{\gamma'(t)}{\gamma(t) - z_0} dt.$$

7.3 Properties of the Winding Number

Proposition 7.4

Let γ be a closed piecewise- C^1 curve. Then $z_0 \mapsto \text{Ind}_{z=z_0}(\gamma)$ is continuous on $\mathbb{C} \setminus R(\gamma)$.

Proof: Fix $z_0 \in \mathbb{C} \setminus R(\gamma)$. Since $R(\gamma)$ is compact, it is closed, so $\mathbb{C} \setminus R(\gamma)$ is open. Thus there is $d > 0$ with $B(z_0, d) \subseteq \mathbb{C} \setminus R(\gamma)$.

If $|z_1 - z_0| < \frac{d}{2}$, then $z_1 \notin R(\gamma)$ and $|z - z_1| \geq \frac{d}{2}$ for every $z \in R(\gamma)$. Hence

$$\left| \frac{1}{z - z_1} - \frac{1}{z - z_0} \right| = \frac{|z_1 - z_0|}{|z - z_1||z - z_0|} \leq \frac{2|z_1 - z_0|}{d^2}.$$

Using the integral formula for the winding number and the ML inequality,

$$|\text{Ind}_{z=z_1}(\gamma) - \text{Ind}_{z=z_0}(\gamma)| \leq \frac{L(\gamma)}{2\pi} \cdot \frac{2|z_1 - z_0|}{d^2}.$$

The right-hand side tends to 0 as $z_1 \rightarrow z_0$, so $\text{Ind}_{z=z_0}(\gamma)$ is continuous on $\mathbb{C} \setminus R(\gamma)$. $\square$

Corollary 7.5

Let γ be a closed piecewise- C^1 curve. Then the function $z_0 \mapsto \text{Ind}_{z=z_0}(\gamma)$ is constant on each connected component of $\mathbb{C} \setminus R(\gamma)$.

Proof: By the proposition, $z_0 \mapsto \text{Ind}_{z=z_0}(\gamma)$ is continuous on each connected component of $\mathbb{C} \setminus R(\gamma)$. It also takes values in $\mathbb{Z}$. The image of a connected set under a continuous map is connected, and the only connected subsets of $\mathbb{Z}$ are singletons. Therefore the winding number is constant on each connected component. $\square$

Proposition 7.6

Let γ be a closed piecewise- C^1 curve. Then $\text{Ind}_{z=z_0}(\gamma) = 0$ for every z_0 in the unbounded connected component of $\mathbb{C} \setminus R(\gamma)$.

Proof: Since $R(\gamma)$ is compact, it is bounded. Choose $M > 0$ such that $|z| \leq M$ for every $z \in R(\gamma)$. If $|z_0| > M + \frac{\ell(\gamma)}{2\pi}$, then $z_0 \notin R(\gamma)$ and

$$|\text{Ind}_{z=z_0}(\gamma)| = \left| \frac{1}{2\pi i} \int_{\gamma} \frac{1}{z - z_0} dz \right| \leq \frac{\ell(\gamma)}{2\pi(|z_0| - M)} < 1.$$

But $\text{Ind}_{z=z_0}(\gamma) \in \mathbb{Z}$, so $\text{Ind}_{z=z_0}(\gamma) = 0$ for such a point z_0 .

This point lies in the unbounded connected component of $\mathbb{C} \setminus R(\gamma)$. By the corollary, the winding number is constant on that component, so it is 0 everywhere on the unbounded component. $\square$

Theorem 7.7**Jordan Curve Theorem**

If $\gamma : [a, b] \rightarrow \mathbb{C}$ is continuous, $\gamma(a) = \gamma(b)$, and γ is injective on $[a, b)$, then $\mathbb{C} \setminus R(\gamma)$ has exactly two connected components.

7.4 Generalized Cauchy Theorem**Definition 7.4**

A *contour* is a possibly empty finite formal sum

$$\Gamma = \sum_{j=1}^m n_j \gamma_j,$$

where $n_j \in \mathbb{Z}$ and each γ_j is a piecewise- C^1 closed curve. Its range is defined to be

$$R(\Gamma) = \cup_{j=1}^m R(\gamma_j).$$

Definition 7.5

If $\Gamma = \sum_{j=1}^m n_j \gamma_j$ is a contour and f is continuous on $R(\Gamma)$, then

$$\int_{\Gamma} f(z) dz = \sum_{j=1}^m n_j \int_{\gamma_j} f(z) dz.$$

Definition 7.6**Winding Number**

If

$$\Gamma = \sum_{j=1}^m n_j \gamma_j$$

is a contour and $z_0 \notin R(\Gamma)$, then

$$\text{Ind}_{z=z_0}(\Gamma) := \sum_{j=1}^m n_j \text{Ind}_{z=z_0}(\gamma_j).$$

Definition 7.7

1. A contour Γ is *simple* if $\text{Ind}_{z=z_0}(\Gamma) \in \{0, 1\}$ for every $z_0 \notin R(\Gamma)$.
2. The *inside* of Γ is

$$\text{int}(\Gamma) := \{z_0 \in \mathbb{C} \setminus R(\Gamma) \mid \text{Ind}_{z=z_0}(\Gamma) = 1\}.$$

Theorem 7.8

Separation Lemma

Suppose $G \subseteq \mathbb{C}$ is open and $K \subseteq G$ is compact. Then there exists a simple contour Γ such that

$$R(\Gamma) \subseteq G \setminus K \quad \text{and} \quad K \subseteq \text{int}(\Gamma).$$

Proof: If $K = \emptyset$, take the empty contour. Now suppose $K \neq \emptyset$. Since K is compact and $\mathbb{C} \setminus G$ is closed, and since $K \subseteq G$, there exists $\delta > 0$ such that

$$|z - w| \geq \delta$$

whenever $z \in K$ and $w \in \mathbb{C} \setminus G$.

Consider the square grid in $\mathbb{C}$ with side length $\eta = \frac{\delta}{3}$. Let $S_1, S_2, \dots, S_p$ be the closed grid squares that intersect K . There are only finitely many such squares because K is bounded. If $z \in S_j$, choose $w \in S_j \cap K$. Then

$$|z - w| \leq \sqrt{2}\eta < \delta,$$

so $z \in G$. Hence

$$U = \cup_{j=1}^p S_j \subseteq G.$$

Regard the selected squares as an integral 2-chain

$$C = \sum_{j=1}^p S_j.$$

Its boundary ∂C is the formal sum of the counter-clockwise boundaries of the selected squares. Every edge shared by two selected squares appears in ∂C twice with opposite orientations, so those interior edges cancel. Thus ∂C consists exactly of the remaining boundary edges of U .

The chain ∂C is a finite union of closed polygonal curves. Let Γ be the contour obtained by taking these boundary curves with their induced orientations. Then $R(\Gamma)$ is exactly the boundary of U .

We claim that $K \cap R(\Gamma) = \emptyset$. Indeed, if $z \in K$ lies on a grid edge, then every grid square adjacent to that edge also intersects K , so that edge is shared by selected squares and cancels. The same argument applies at grid vertices. Therefore every point of K lies in the interior of U , and no point of K lies on $R(\Gamma)$. Since $U \subseteq G$, this gives $R(\Gamma) \subseteq G \setminus K$.

Finally, the winding number of the positively oriented boundary of a square is 1 inside the square and 0 outside the square. By additivity of winding number over contours, the cancellations above imply that $\text{Ind}_{z=z_0}(\Gamma)$ is 1 on the interior of U and 0 outside U for every $z_0 \notin R(\Gamma)$. Hence Γ is simple, and since K lies in the interior of U , we have $K \subseteq \text{int}(\Gamma)$.

Note: See relation to 1-chain, homology. □

Corollary 7.9

If we have Γ as constructed in the Separation Lemma, and if $f : G \rightarrow \mathbb{C}$ is holomorphic, then for $z_0 \in K$,

$$f(z_0) = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(z)}{z - z_0} dz$$

Proof: In the proof of the Separation Lemma, Γ is the boundary of a finite union of closed squares contained in G . Equivalently, Γ is obtained from the formal sum of the positively oriented boundaries of those squares after cancelling shared edges.

Let h be holomorphic on G . By Cauchy's theorem for convex regions, $\int_{\partial S_j} h(z) dz = 0$ for each selected square S_j . Summing over the selected squares and using the edge cancellations gives

$$\int_{\Gamma} h(z) dz = 0.$$

Fix $z_0 \in K$ and define

$$h(z) = \begin{cases} \frac{f(z) - f(z_0)}{z - z_0}, & z \neq z_0 \\ f'(z_0), & z = z_0 \end{cases}$$

The singularity at z_0 is removable, so h is holomorphic on G . Therefore

$$0 = \int_{\Gamma} h(z) dz = \int_{\Gamma} \frac{f(z)}{z - z_0} dz - f(z_0) \int_{\Gamma} \frac{1}{z - z_0} dz.$$

Since $K \subseteq \text{int}(\Gamma)$, we have $\text{Ind}_{z=z_0}(\Gamma) = 1$, and hence

$$\int_{\Gamma} \frac{1}{z - z_0} dz = 2\pi i.$$

Thus

$$f(z_0) = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(z)}{z - z_0} dz.$$

□

Theorem 7.10**Cauchy's Theorem**

Let G be an open set, let Γ be a contour with $R(\Gamma) \subseteq G$, and suppose

$$\text{Ind}_{z=w}(\Gamma) = 0$$

for every $w \in \mathbb{C} \setminus G$. If $f : G \rightarrow \mathbb{C}$ is holomorphic, then

$$\int_{\Gamma} f(z) dz = 0$$

Proof: Set

$$K = R(\Gamma) \cup \{z \in \mathbb{C} \setminus R(\Gamma) \mid \text{Ind}_{z=z}(\Gamma) \neq 0\}.$$

The winding number is locally constant on $\mathbb{C} \setminus R(\Gamma)$ and is 0 on the unbounded component of $\mathbb{C} \setminus R(\Gamma)$, so the second set is a bounded union of connected components of $\mathbb{C} \setminus R(\Gamma)$. Hence K is compact. By the hypothesis on winding numbers outside G , we have $K \subseteq G$.

By the Separation Lemma, there is a simple contour Γ' such that

$$R(\Gamma') \subseteq G \setminus K \quad \text{and} \quad K \subseteq \text{int}(\Gamma').$$

Applying the preceding corollary to this separating contour, for every $z \in K$ we have

$$f(z) = \frac{1}{2\pi i} \int_{\Gamma'} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

In particular, this formula holds for every $z \in R(\Gamma)$. Therefore

$$\int_{\Gamma} f(z) dz = \frac{1}{2\pi i} \int_{\Gamma} \left(\int_{\Gamma'} \frac{f(\zeta)}{\zeta - z} d\zeta \right) dz.$$

Since both contours are finite sums of piecewise- C^1 curves and the integrand is continuous on $R(\Gamma') \times R(\Gamma)$, we may interchange the two integrals:

$$\int_{\Gamma} f(z) dz = \frac{1}{2\pi i} \int_{\Gamma'} f(\zeta) \left(\int_{\Gamma} \frac{1}{\zeta - z} dz \right) d\zeta.$$

For $\zeta \in R(\Gamma')$, we have $\zeta \notin K$, so $\zeta \notin R(\Gamma)$ and $\text{Ind}_{z=\zeta}(\Gamma) = 0$. Hence

$$\int_{\Gamma} \frac{1}{\zeta - z} dz = - \int_{\Gamma} \frac{1}{z - \zeta} dz = -2\pi i \text{Ind}_{z=\zeta}(\Gamma) = 0.$$

Substituting this into the preceding expression gives

$$\int_{\Gamma} f(z) dz = 0.$$

□

Note: We now see that our old results easily follow from this generalized result:

Corollary 7.11

Cauchy's Theorem For Convex Regions

G convex, $f : G \rightarrow \mathbb{C}$ holomorphic, Γ piecewise- C^1 in G . So $\text{int}(\gamma) \subseteq G$. Therefore, $\int f(z) dz = 0$.

Corollary 7.12

Cauchy's Theorem for Annulus

It follows from Generalized Cauchy's Theorem.

7.5 Residues and the Residue Theorem

Definition 7.8

Suppose f is holomorphic with an isolated singularity $z_0 \in C$. Denote the Laurent series of f as $\sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$. We define the residue of f to be a_{-1} and denote it as $\text{res}_{z=z_0}(f(z))$, $\text{res}(f, z_0)$, or $\text{res}_{z_0}(f)$.

Proposition 7.13

Suppose g and h are holomorphic near z_0 , with

$$h(z_0) = 0, \quad h'(z_0) \neq 0, \quad \text{and} \quad g(z_0) \neq 0.$$

Then $\frac{g}{h}$ has a simple pole at z_0 .

Proof: Since $h(z_0) = 0$ and $h'(z_0) \neq 0$, the function h has a simple zero at z_0 . Thus we may write

$$h(z) = (z - z_0)k(z),$$

where k is holomorphic near z_0 and $k(z_0) = h'(z_0) \neq 0$. Therefore

$$\frac{g(z)}{h(z)} = \frac{1}{z - z_0} \cdot \frac{g(z)}{k(z)}.$$

Since $k(z_0) \neq 0$, the function $\frac{g}{k}$ is holomorphic near z_0 . Also $(\frac{g}{k})(z_0) \neq 0$ because $g(z_0) \neq 0$. Hence $\frac{g}{h}$ has a simple pole at z_0 . □

Theorem 7.14

Residue at a Pole

Suppose z_0 is a pole of order m of f . Then

$$\text{res}_{z=z_0}(f) = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} ((z - z_0)^m f(z)).$$

Proof: Since z_0 is a pole of order m , we may write

$$f(z) = \sum_{n=-m}^{\infty} a_n (z - z_0)^n$$

near z_0 , with $a_{-m} \neq 0$. Hence

$$(z - z_0)^m f(z) = \sum_{n=0}^{\infty} a_{n-m} (z - z_0)^n.$$

The coefficient of $(z - z_0)^{m-1}$ in this Taylor series is a_{-1} . Therefore

$$a_{-1} = \frac{1}{(m-1)!} \lim_{z \rightarrow z_0} \frac{d^{m-1}}{dz^{m-1}} ((z - z_0)^m f(z)).$$

Since $\text{res}_{z=z_0}(f) = a_{-1}$, the formula follows. $\square$

Corollary 7.15

Residue at a Simple Pole

Suppose z_0 is a simple pole of f . Then

$$\text{res}_{z=z_0}(f) = \lim_{z \rightarrow z_0} (z - z_0) f(z).$$

Proof: This is the previous theorem with $m = 1$. $\square$

Exercise 7.5.1

Compute the residues of the following functions at each of their isolated singularities:

1. $e^{\frac{1}{z}}$
 2. $\frac{z^3}{z-1}$
 3. $\frac{z^3}{(z-1)^3}$
 4. $\frac{z^k}{(z-a)(z-b)^2}$
- for $k \geq 0$, $a \neq b$.
5. $\frac{\sin z}{2z^2 - 5z + 2}$
 6. $\frac{2z^2 - 5z + 2}{\sin z}$

Note

$$a_n = \frac{1}{2\pi i} \int_{C_r} (z - z_0)^{-n-1} f(z) dz \text{ and } \int_{C_r} f(z) dz = 2\pi i \text{res}_{z_0} f.$$

Theorem 7.16**Residue Theorem**

Let $G \subseteq \mathbb{C}$ be open, and let f be holomorphic on $G \setminus \{z_1, \dots, z_p\}$, where $z_1, \dots, z_p$ are isolated singularities of f . Let Γ be a contour such that

$$R(\Gamma) \subseteq G \setminus \{z_1, \dots, z_p\} \quad \text{and} \quad \text{Ind}_{z=w}(\Gamma) = 0 \quad \text{for every } w \in \mathbb{C} \setminus G.$$

Then

$$\int_{\Gamma} f(z) \, dz = 2\pi i \sum_{j=1}^p \text{Ind}_{z=z_j}(\Gamma) \text{res}_{z=z_j}(f).$$

Proof: Choose $r > 0$ so small that the closed disks $\overline{D(z_j, r)}$ are pairwise disjoint, lie in G , and do not meet $R(\Gamma)$. Let C_j be the positively oriented circle $C_{r(z_j)} = \partial D(z_j, r)$, and set

$$n_j = \text{Ind}_{z=z_j}(\Gamma) \quad \text{and} \quad \Lambda = \Gamma - \sum_{j=1}^p n_j C_j.$$

We apply Cauchy's theorem to Λ in $U = G \setminus \{z_1, \dots, z_p\}$. If $w \notin G$, then each C_j has winding number 0 about w , so $\text{Ind}_{z=w}(\Lambda) = \text{Ind}_{z=w}(\Gamma) = 0$. Also, for each k ,

$$\text{Ind}_{z=z_k}(\Lambda) = \text{Ind}_{z=z_k}(\Gamma) - n_k = 0,$$

since $\text{Ind}_{z=z_k}(C_j) = 1$ when $j = k$ and 0 otherwise. Thus $\text{Ind}_{z=w}(\Lambda) = 0$ for every $w \in \mathbb{C} \setminus U$. Since f is holomorphic on U , Cauchy's theorem gives

$$0 = \int_{\Lambda} f(z) \, dz = \int_{\Gamma} f(z) \, dz - \sum_{j=1}^p n_j \int_{C_j} f(z) \, dz.$$

For each j , write the Laurent expansion of f near z_j as

$$f(z) = \sum_{n=-\infty}^{\infty} a_{j,n} (z - z_j)^n.$$

This series converges uniformly on C_j , so we may integrate term by term. The integral of $(z - z_j)^n$ around C_j is 0 unless $n = -1$, and it is $2\pi i$ when $n = -1$. Hence

$$\int_{C_j} f(z) \, dz = 2\pi i a_{j,-1} = 2\pi i \text{res}_{z=z_j}(f).$$

Substituting this into the previous equality gives

$$\int_{\Gamma} f(z) \, dz = 2\pi i \sum_{j=1}^p \text{Ind}_{z=z_j}(\Gamma) \text{res}_{z=z_j}(f).$$

□

Corollary 7.17**Cauchy's Formula**

If Γ is a simple contour with interior in G and if $f : G \rightarrow \mathbb{C}$ is holomorphic, then $f(z) = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - z} d\zeta$ for all $z \in \text{int}(\Gamma)$.

7.6 Applications of the Residue Theorem**Exercise 7.6.1**

Compute

$$\int_{C_1} (z-2)^{-1} (2z+1)^{-2} (3z-1)^{-3} dz$$

Exercise 7.6.2

Compute $\int_0^\infty \frac{\sin x}{x} dx$. Hint: Consider $f(z) = \frac{e^{iz}}{z}$.

Check that imaginary part is $\frac{\sin x}{x}$.

Exercise 7.6.3

Compute

$$\int_0^\infty \frac{1}{1+z^a} dx$$

for $a > 1$

Proof: Define $f(z) = \frac{1}{1+z^a}$, where z^a is a branch such that $1^a = 1$. □

7.7 Simply Connected Regions**Definition 7.9****Homotopy of Paths**

Let $G \subseteq \mathbb{C}$, and let $\gamma_0, \gamma_1 : [0, 1] \rightarrow G$ be paths with the same initial point z_0 and terminal point z_1 . A *homotopy from γ_0 to γ_1 in G relative to their endpoints* is a continuous map

$$H : [0, 1]^2 \rightarrow G$$

such that, for every $s, t \in [0, 1]$,

$$H(s, 0) = \gamma_0(s), \quad H(s, 1) = \gamma_1(s),$$

and

$$H(0, t) = z_0, \quad H(1, t) = z_1.$$

If such an H exists, then γ_0 and γ_1 are *homotopic in G relative to their endpoints*.

Definition 7.10**Null-Homotopic Closed Curve**

Let $\gamma : [0, 1] \rightarrow G$ be a closed curve based at $z_0 = \gamma(0) = \gamma(1)$. Then γ is *null-homotopic in G* if it is homotopic relative to its endpoints to the constant curve

$$c_{z_0}(s) = z_0.$$

Equivalently, γ represents the identity element $[c_{z_0}]$ of the fundamental group $\pi_1(G, z_0)$.

Definition 7.11**One-Point Compactification**

We write

$$\hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$$

and equip it with the *one-point compactification topology*. The usual open subsets of $\mathbb{C}$ remain open, while a neighbourhood of ∞ is any set containing a set of the form

$$\{\infty\} \cup (\mathbb{C} \setminus K),$$

where $K \subseteq \mathbb{C}$ is compact. Thus a path $\gamma : [0, 1] \rightarrow \hat{\mathbb{C}}$ with $\gamma(1) = \infty$ is continuous at 1 whenever

$$|\gamma(t)| \rightarrow \infty \quad \text{as } t \rightarrow 1^-.$$

Definition 7.12**Simply Connected**

Let $G \subseteq \mathbb{C}$ be a region. The *extended complement* of G is

$$\hat{\mathbb{C}} \setminus G.$$

We say that G is *simply connected* if its extended complement is connected.

Lemma 7.18

Let $G \subseteq \mathbb{C}$ be open and connected. If $z_0, z \in G$, then there is a piecewise- $\mathcal{C}^1$ curve $\gamma : [0, 1] \rightarrow G$ such that

$$\gamma(0) = z_0 \quad \text{and} \quad \gamma(1) = z.$$

Proof: Fix $z_0 \in G$, and let A be the set of points in G that can be joined to z_0 by a polygonal path in G . Clearly $z_0 \in A$. If $z \in A$, choose $r > 0$ such that $D(z, r) \subseteq G$. Every point of $D(z, r)$ can be joined to z by a line segment in $D(z, r)$, and hence to z_0 by a polygonal path in G . Thus $D(z, r) \subseteq A$, so A is open in G .

If $z \in G \setminus A$, choose $r > 0$ such that $D(z, r) \subseteq G$. If some $w \in D(z, r)$ belonged to A , adjoining the line segment from w to z would give a polygonal path from z_0 to z , a

contradiction. Therefore $D(z, r) \subseteq G \setminus A$, so $G \setminus A$ is also open in G . Since G is connected and A is nonempty, open, and closed in G , we have $A = G$. A polygonal path is piecewise- C^1 , which proves the result. $\square$

Example

Set

$$S_+ = \{z \in \mathbb{C} \mid \operatorname{Re}(z) > 1\} \quad \text{and} \quad S_- = \{z \in \mathbb{C} \mid \operatorname{Re}(z) < -1\},$$

and let

$$S = S_+ \cup S_- \cup \{\infty\}.$$

We show that S is path-connected. For $z \in S_+$, define

$$\gamma_z^+(t) = \begin{cases} z + \frac{t}{1-t} & 0 \leq t < 1 \\ \infty & t = 1. \end{cases}$$

For $t < 1$, we have $\operatorname{Re}(\gamma_z^+(t)) > 1$, so the path remains in S_+ , and

$$|\gamma_z^+(t)| \rightarrow \infty \quad \text{as} \quad t \rightarrow 1^-.$$

By the definition of the one-point compactification topology, γ_z^+ is continuous at 1. Hence it is a path in $S_+ \cup \{\infty\}$ from z to ∞ .

Similarly, for $z \in S_-$, the map

$$\gamma_z^-(t) = \begin{cases} z - \frac{t}{1-t} & 0 \leq t < 1 \\ \infty & t = 1 \end{cases}$$

is a path in $S_- \cup \{\infty\}$ from z to ∞ . Thus every point of S can be joined to ∞ by a path in S , so S is path-connected and hence connected. Notice that $S \setminus \{\infty\} = S_+ \cup S_-$ is disconnected in the usual topology on $\mathbb{C}$.

Theorem 7.19

Characterizations of Simply Connected Regions

Let $G \subseteq \mathbb{C}$ be a region. The following are equivalent:

1. G is simply connected.
2. If Γ is a contour in G , then

$$\operatorname{Ind}_{z=w}(\Gamma) = 0$$

for every $w \in \mathbb{C} \setminus G$.

3. If Γ is a contour in G , and $f : G \rightarrow \mathbb{C}$ is holomorphic, then $\int_{\Gamma} f(z) dz = 0$.
4. If $f : G \rightarrow \mathbb{C}$ is holomorphic, then f has a primitive.
5. If $f : G \rightarrow \mathbb{C} \setminus \{0\}$ is holomorphic, then there is a branch of $\log f$.
6. Every closed curve in G is null-homotopic in G .

Note

The implications

$$(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (4) \Rightarrow (5) \Rightarrow (1)$$

are proved directly below. The Riemann Mapping Theorem gives $(1) \Rightarrow (6)$, while $(6) \Rightarrow (5)$ follows from the lifting property of the covering map $\exp : \mathbb{C} \rightarrow \mathbb{C}^*$.

Proof of Characterizations of Simply Connected Regions: We first prove the cycle of implications among (1) – (5).

(1) $\Rightarrow$ (2): Let Γ be a contour in G , and set

$$A = \hat{\mathbb{C}} \setminus G.$$

Since $R(\Gamma) \subseteq G$, we have

$$A \subseteq \hat{\mathbb{C}} \setminus R(\Gamma).$$

By (1), A is connected, and $\infty \in A$. Therefore A is contained in the connected component of $\hat{\mathbb{C}} \setminus R(\Gamma)$ containing ∞ . Equivalently, every point of $\mathbb{C} \setminus G$ lies in the unbounded component of $\mathbb{C} \setminus R(\Gamma)$. The winding number of Γ is 0 on this component, so

$$\text{Ind}_{z=w}(\Gamma) = 0$$

for every $w \in \mathbb{C} \setminus G$, so condition (2) holds.

(2) $\Rightarrow$ (3): Let Γ be a contour in G , and let $f : G \rightarrow \mathbb{C}$ be holomorphic. By (2), $\text{Ind}_{z=w}(\Gamma) = 0$ for every $w \in \mathbb{C} \setminus G$. Since $R(\Gamma) \subseteq G$, the generalized Cauchy theorem gives

$$\int_{\Gamma} f(z) dz = 0.$$

(3) $\Rightarrow$ (4): Let $f : G \rightarrow \mathbb{C}$ be holomorphic, and fix $z_0 \in G$. By the lemma, for each $z \in G$ there is a piecewise- C^1 curve γ_z in G from z_0 to z . Define

$$F(z) = \int_{\gamma_z} f(\zeta) d\zeta.$$

This is well-defined. Indeed, if γ_1 and γ_2 are two such curves, traverse γ_1 and then traverse γ_2 in reverse. The resulting curve Γ is a closed piecewise- C^1 curve in G , so by (3),

$$0 = \int_{\Gamma} f(\zeta) d\zeta = \int_{\gamma_1} f(\zeta) d\zeta - \int_{\gamma_2} f(\zeta) d\zeta.$$

Hence both choices give the same value of $F(z)$.

We now show that $F' = f$. Fix $z \in G$, and choose $r > 0$ such that $D(z, r) \subseteq G$. If $0 < |h| < r$, use a curve from z_0 to z followed by the line segment $\sigma_{h(t)} = z + th$, $0 \leq t \leq 1$, to compute $F(z + h)$. By well-definedness,

$$\frac{F(z + h) - F(z)}{h} = \frac{1}{h} \int_{\sigma_h} f(\zeta) d\zeta = \int_0^1 f(z + th) dt.$$

Therefore,

$$\left| \frac{F(z + h) - F(z)}{h} - f(z) \right| \leq \int_0^1 |f(z + th) - f(z)| dt.$$

The right-hand side tends to 0 as $h \rightarrow 0$ because f is continuous at z . Thus $F'(z) = f(z)$. Since $z \in G$ was arbitrary, F is a primitive of f on G .

(4) $\Rightarrow$ (5): Let $f : G \rightarrow \mathbb{C} \setminus \{0\}$ be holomorphic. Since f never vanishes, the function

$$h = \frac{f'}{f}$$

is holomorphic on G . By (4), there is a holomorphic function $H : G \rightarrow \mathbb{C}$ such that

$$H' = h = \frac{f'}{f}.$$

Consider the holomorphic function fe^{-H} . By the product and chain rules,

$$(fe^{-H})' = f'e^{-H} - fH'e^{-H} = e^{-H} \left(f' - f \frac{f'}{f} \right) = 0.$$

Since G is connected, fe^{-H} is constant on G . Thus there is some $c \in \mathbb{C} \setminus \{0\}$ such that

$$f(z)e^{-H(z)} = c$$

for every $z \in G$. Choose $a \in \mathbb{C}$ such that $e^a = c$, and define

$$L = H + a.$$

Then L is holomorphic on G and

$$e^{L(z)} = e^{H(z)}e^a = ce^{H(z)} = f(z)$$

for every $z \in G$. Therefore L is a branch of $\log f$ on G .

(5) $\Rightarrow$ (1): Suppose, toward a contradiction, that

$$K = \hat{\mathbb{C}} \setminus G$$

is disconnected. Since K is compact, there are disjoint nonempty compact sets A and B such that

$$K = A \cup B.$$

Relabel them if necessary so that $\infty \in B$. Set

$$U = \hat{\mathbb{C}} \setminus B.$$

Since $\infty \in B$, the set U is an open subset of $\mathbb{C}$, and A is a compact subset of U . By the Separation Lemma, there is a simple contour Γ such that

$$R(\Gamma) \subseteq U \setminus A = G \quad \text{and} \quad A \subseteq \text{int}(\Gamma).$$

Choose $w_0 \in A$, and define

$$f : G \rightarrow \mathbb{C} \setminus \{0\}, \quad f(z) = z - w_0.$$

This function is holomorphic and never vanishes on G because $w_0 \notin G$. By (5), it has a holomorphic branch of logarithm $L : G \rightarrow \mathbb{C}$, so

$$e^{L(z)} = z - w_0.$$

Differentiating gives

$$L'(z)e^{L(z)} = 1,$$

and hence

$$L'(z) = \frac{1}{z - w_0}.$$

Since L is a primitive of $\frac{1}{z - w_0}$ on G and Γ is a contour in G , the fundamental theorem of calculus gives

$$\int_{\Gamma} \frac{1}{z - w_0} dz = 0.$$

On the other hand, $w_0 \in A \subseteq \text{int}(\Gamma)$, so $\text{Ind}_{z=w_0}(\Gamma) = 1$. Therefore,

$$\int_{\Gamma} \frac{1}{z - w_0} dz = 2\pi i \text{Ind}_{z=w_0}(\Gamma) = 2\pi i,$$

a contradiction. Thus $\hat{\mathbb{C}} \setminus G$ is connected.

(1) $\Rightarrow$ (6): Let $\gamma : [0, 1] \rightarrow G$ be a closed curve. By the Riemann Mapping Theorem, either $G = \mathbb{C}$ or there is a biholomorphism $\varphi : G \rightarrow \mathbb{D}$.

If $G = \mathbb{C}$, define

$$H(s, t) = (1 - t)\gamma(s) + t\gamma(0).$$

Since $\mathbb{C}$ is convex, H is a null-homotopy of γ in $\mathbb{C}$.

Otherwise, define

$$K(s, t) = (1 - t)(\varphi \circ \gamma)(s) + t(\varphi \circ \gamma)(0).$$

Since $\mathbb{D}$ is convex, K is a null-homotopy of $\varphi \circ \gamma$ in $\mathbb{D}$. Therefore

$$H = \varphi^{-1} \circ K$$

is a null-homotopy of γ in G . Hence every closed curve in G is null-homotopic in G .

(6) $\Rightarrow$ (5): Let $f : G \rightarrow \mathbb{C}^*$ be holomorphic. Fix $z_0 \in G$ and choose $a \in \mathbb{C}$ such that $e^a = f(z_0)$. By (6), every closed curve in G is null-homotopic, so its image under f is null-homotopic in $\mathbb{C}^*$. The lifting property of the covering map

$$\exp : \mathbb{C} \rightarrow \mathbb{C}^*$$

therefore gives a continuous lift $L : G \rightarrow \mathbb{C}$ such that

$$L(z_0) = a \quad \text{and} \quad e^{L(z)} = f(z)$$

for every $z \in G$.

The lift L is holomorphic. Indeed, near each point of G , the image of f lies in a small disc in $\mathbb{C}^*$ on which there is a holomorphic branch ℓ of logarithm. On a sufficiently small connected neighbourhood,

$$L = (\ell \circ f)$$

is continuous and takes values in $2\pi i\mathbb{Z}$, so it is constant. Thus L differs locally from the holomorphic function $\ell \circ f$ by a constant. Therefore L is holomorphic on G and is a branch of $\log f$. □

Theorem 7.20

Riemann Mapping Theorem

Let $G \subseteq \mathbb{C}$ be a nonempty region such that $\hat{\mathbb{C}} \setminus G$ is connected. Then either $G = \mathbb{C}$ or there exists a biholomorphism

$$\varphi : G \rightarrow \mathbb{D}.$$

That is, φ is a holomorphic bijection whose inverse is also holomorphic.

Proof: Out of scope. □

8 Linear-Fractional Transformations

8.1 The Riemann Sphere and the Projective Line

Definition 8.1

Riemann Sphere

The **Riemann sphere** is

$$\hat{\mathbb{C}} := \mathbb{C} \cup \{\infty\}.$$

It is equipped with the one-point compactification topology: the usual open subsets of $\mathbb{C}$ are open in $\hat{\mathbb{C}}$, and a neighbourhood of ∞ is any set containing a set of the form

$$\{\infty\} \cup (\mathbb{C} \setminus K),$$

where $K \subseteq \mathbb{C}$ is compact.

Definition 8.2

Projective Line

Let $\mathbb{F}$ be either $\mathbb{R}$ or $\mathbb{C}$. On $\mathbb{F}^2 \setminus \{(0, 0)\}$, define

$$(x_0, x_1) \sim (y_0, y_1) \Leftrightarrow (y_0, y_1) = \lambda(x_0, x_1)$$

for some $\lambda \in \mathbb{F} \setminus \{0\}$. The **projective line over $\mathbb{F}$** is the quotient

$$\mathbb{F}\mathbb{P}^1 = (\mathbb{F}^2 \setminus \{(0, 0)\}) / \sim.$$

The equivalence class of (x_0, x_1) is denoted by $[x_0 : x_1]$. These are called **homogeneous coordinates** because

$$[x_0 : x_1] = [\lambda x_0 : \lambda x_1]$$

for every $\lambda \in \mathbb{F} \setminus \{0\}$. Equivalently, $\mathbb{F}\mathbb{P}^1$ is the set of one-dimensional $\mathbb{F}$ -linear subspaces of $\mathbb{F}^2$.

Note

In particular, $\mathbb{R}\mathbb{P}^1$ is the space of lines through the origin in $\mathbb{R}^2$. Moreover,

$$\mathbb{R}\mathbb{P}^1 \cong S^1 \cong \hat{\mathbb{R}},$$

where $\hat{\mathbb{R}} = \mathbb{R} \cup \{\infty\}$ is the one-point compactification of $\mathbb{R}$.

Proposition 8.1

The map

$$\iota : \hat{\mathbb{C}} \rightarrow \mathbb{CP}^1, \quad \iota(z) = [z : 1], \quad \iota(\infty) = [1 : 0],$$

is a bijection. When $\mathbb{CP}^1$ has the quotient topology, ι is a homeomorphism.

Proof: Every point of $\mathbb{CP}^1$ has the form $[z_0 : z_1]$. If $z_1 \neq 0$, then

$$[z_0 : z_1] = \left[\frac{z_0}{z_1} : 1 \right] = \iota \left(\frac{z_0}{z_1} \right).$$

If $z_1 = 0$, then $z_0 \neq 0$ and $[z_0 : 0] = [1 : 0] = \iota(\infty)$. These two forms are unique, so ι is bijective.

On the chart

$$U_1 = \{[z_0 : z_1] : z_1 \neq 0\},$$

the coordinate map $[z_0 : z_1] \rightarrow \frac{z_0}{z_1}$ identifies U_1 with $\mathbb{C}$. On the chart

$U_0 = \{[z_0 : z_1] : z_0 \neq 0\}$, the coordinate map $[z_0 : z_1] \rightarrow \frac{z_1}{z_0}$ identifies a neighbourhood of $[1 : 0]$ with a neighbourhood of 0. Under ι^{-1} , the set where $\left| \frac{z_1}{z_0} \right| < \varepsilon$ corresponds to

$$\{\infty\} \cup \left\{ z \in \mathbb{C} : |z| > \frac{1}{\varepsilon} \right\},$$

which is a neighbourhood of ∞ in the one-point compactification topology. Thus ι and ι^{-1} are continuous. □

We use this homeomorphism to identify $\mathbb{CP}^1$ with $\hat{\mathbb{C}}$.

8.2 From Matrices to Linear-Fractional Transformations

Proposition 8.2

Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in GL_2(\mathbb{C}).$$

Then A induces a well-defined bijection

$$\Phi_A : \mathbb{CP}^1 \rightarrow \mathbb{CP}^1, \quad \Phi_A([v]) = [Av].$$

Under the identification $\mathbb{CP}^1 \cong \hat{\mathbb{C}}$, this map is given by

$$\Phi_A(z) = \begin{cases} \frac{az+b}{cz+d} & \text{if } z \in \mathbb{C} \text{ and } cz+d \neq 0 \\ \infty & \text{if } z \in \mathbb{C} \text{ and } cz+d = 0 \\ \frac{a}{c} & \text{if } z = \infty \text{ and } c \neq 0 \\ \infty & \text{if } z = \infty \text{ and } c = 0. \end{cases}$$

Proof: If $[v] = [\lambda v]$ for $\lambda \neq 0$, then

$$[A(\lambda v)] = [\lambda Av] = [Av],$$

so the map is independent of the representative. Since A is invertible, $Av \neq 0$ whenever $v \neq 0$, and $\Phi_{A^{-1}}$ is the inverse of Φ_A .

For $z \in \mathbb{C}$,

$$\Phi_{A([z:1])} = [az + b : cz + d].$$

If $cz + d \neq 0$, this is $\left[\frac{az+b}{cz+d} : 1\right]$. If $cz + d = 0$, then $az + b \neq 0$, since otherwise $A(z, 1)^T = (0, 0)^T$, contradicting the invertibility of A ; hence the image is $[1 : 0]$. Finally,

$$\Phi_{A([1:0])} = [a : c],$$

which gives the two stated cases at ∞ . □

Definition 8.3

Linear-Fractional Transformation

A **linear-fractional transformation**, also called a **Möbius transformation**, is a map $\varphi : \hat{\mathbb{C}} \rightarrow \hat{\mathbb{C}}$ of the form $\varphi = \Phi_A$ for some $A \in GL_2(\mathbb{C})$. Thus, away from its possible pole,

$$\varphi(z) = \frac{az + b}{cz + d}, \quad ad - bc \neq 0,$$

with the values at the pole and at ∞ defined as in the preceding proposition.

Note

If $c \neq 0$, then $-\frac{d}{c}$ is the unique pole of φ and

$$\varphi\left(-\frac{d}{c}\right) = \infty, \quad \varphi(\infty) = \frac{a}{c}.$$

If $c = 0$, then φ is the affine map

$$\varphi(z) = \frac{a}{d}z + \frac{b}{d}$$

on $\mathbb{C}$ and fixes ∞ .

Theorem 8.3

Group of Linear-Fractional Transformations

The linear-fractional transformations form a group under composition. The map

$$\rho : GL_2(\mathbb{C}) \rightarrow \text{Möb}(\hat{\mathbb{C}}), \quad \rho(A) = \Phi_A,$$

is a surjective group homomorphism with kernel

$$\ker(\rho) = \{\lambda I_2 : \lambda \in \mathbb{C} \setminus \{0\}\}.$$

Consequently,

$$\text{Möb}(\hat{\mathbb{C}}) \cong GL_2(\mathbb{C}) / \{\lambda I_2 : \lambda \in \mathbb{C} \setminus \{0\}\},$$

and this quotient is denoted by $PGL_2(\mathbb{C})$.

Proof: For $A, B \in GL_2(\mathbb{C})$ and $[v] \in \mathbb{CP}^1$,

$$(\Phi_A \circ \Phi_B)([v]) = \Phi_{A([Bv])} = [ABv] = \Phi_{AB}([v]).$$

Therefore $\rho(AB) = \rho(A) \circ \rho(B)$. Surjectivity follows from the definition of a linear-fractional transformation.

If $A = \lambda I_2$ with $\lambda \neq 0$, then $[Av] = [v]$ for every nonzero v , so $A \in \ker(\rho)$. Conversely, suppose Φ_A is the identity. Write $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Since Φ_A fixes $[1 : 0]$ and $[0 : 1]$, we must have $c = 0$ and $b = 0$. Since it also fixes $[1 : 1]$, we have $[a : d] = [1 : 1]$, and hence $a = d$. Thus $A = aI_2$, with $a \neq 0$ because A is invertible. The claimed isomorphism now follows from the First Isomorphism Theorem. $\square$

Corollary 8.4

If $A, B \in GL_2(\mathbb{C})$, then $\Phi_A = \Phi_B$ if and only if

$$A = \lambda B$$

for some $\lambda \in \mathbb{C} \setminus \{0\}$.

Proof: We have $\Phi_A = \Phi_B$ exactly when $\Phi_{B^{-1}A}$ is the identity. By the kernel computation in the preceding theorem, this holds exactly when $B^{-1}A = \lambda I_2$ for some $\lambda \neq 0$, or equivalently when $A = \lambda B$. $\square$

Note

Over $\mathbb{C}$, this group may also be written as

$$PSL_2(\mathbb{C}) = SL_2(\mathbb{C})/\{\pm I_2\}.$$

Indeed, if $A \in GL_2(\mathbb{C})$, choose $\mu \in \mathbb{C} \setminus \{0\}$ such that $\mu^2 = \det(A)^{-1}$. Then $\mu A \in SL_2(\mathbb{C})$ and induces the same linear-fractional transformation as A . The two choices of μ differ by a sign, which accounts for the quotient by $\{\pm I_2\}$.

Corollary 8.5

If $\varphi = \Phi_A$ with $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then

$$\varphi^{-1}(z) = \frac{dz - b}{-cz + a},$$

with the usual conventions at its pole and at ∞ .

Proof: Since scalar multiples induce the same projective map, the factor $\frac{1}{ad-bc}$ in

$$A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

may be omitted. The result follows from $\Phi_A^{-1} = \Phi_{A^{-1}}$. $\square$

8.3 Analytic and Geometric Properties

Proposition 8.6

Every linear-fractional transformation is a biholomorphism of the Riemann sphere. At each finite point other than its pole,

$$\varphi'(z) = \frac{ad-bc}{(cz+d)^2} \neq 0.$$

Proof: The derivative formula follows from the quotient rule. At the pole $z_0 = -\frac{d}{c}$, use the target coordinate $w \rightarrow \frac{1}{w}$ about ∞ ; in this coordinate the map is

$$\frac{1}{\varphi(z)} = \frac{cz+d}{az+b},$$

which is holomorphic near z_0 because $az_0 + b \neq 0$.

To check holomorphicity at ∞ , use the source coordinate $w = \frac{1}{z}$. If $c \neq 0$, then

$$\varphi\left(\frac{1}{w}\right) = \frac{a + bw}{c + dw}$$

is holomorphic near $w = 0$. If $c = 0$, then $\varphi(\infty) = \infty$, and in the target coordinate at ∞ we have

$$\frac{1}{\varphi\left(\frac{1}{w}\right)} = \frac{dw}{a + bw},$$

which is holomorphic near $w = 0$. Hence φ is holomorphic everywhere on $\hat{\mathbb{C}}$. Its inverse is also a linear-fractional transformation by the preceding corollary, so φ is biholomorphic. $\square$

Proposition 8.7

Three-Point Property

Given two ordered triples of distinct points

$$z_1, z_2, z_3 \in \hat{\mathbb{C}} \quad \text{and} \quad w_1, w_2, w_3 \in \hat{\mathbb{C}},$$

there is a unique linear-fractional transformation φ such that

$$\varphi(z_j) = w_j$$

for $j = 1, 2, 3$.

Proof: Regard the points as lines in $\mathbb{C}^2$. Choose nonzero representatives v_1 and v_3 of z_1 and z_3 . Since these points are distinct, v_1, v_3 is a basis. A representative of z_2 therefore has the form $\alpha v_1 + \beta v_3$, where $\alpha, \beta \neq 0$; after rescaling v_1 and v_3 , we may take $v_1 + v_3$ as a representative of z_2 . Similarly, choose representatives u_1, u_3 of w_1, w_3 so that $u_1 + u_3$ represents w_2 . The unique linear map A satisfying

$$Av_1 = u_1, \quad Av_3 = u_3$$

is invertible and induces a linear-fractional transformation with the desired values.

For uniqueness, suppose Φ_A and Φ_B both have the desired values. Then $\Phi_{B^{-1}A}$ fixes the three distinct lines represented by v_1, v_3 , and $v_1 + v_3$. Fixing the first two lines makes the matrix of $B^{-1}A$ diagonal in the basis v_1, v_3 ; fixing the third forces its two diagonal entries to agree. Thus $B^{-1}A$ is scalar, so $\Phi_A = \Phi_B$. $\square$

Corollary 8.8

A linear-fractional transformation that fixes three distinct points of $\hat{\mathbb{C}}$ is the identity.

Proposition 8.9

Fixed Points

A nonidentity linear-fractional transformation has exactly one or two fixed points in $\hat{\mathbb{C}}$.

Proof: Let $\varphi = \Phi_A$. Under the identification $\hat{\mathbb{C}} \cong \mathbb{CP}^1$, a point $[v]$ is fixed precisely when

$$[Av] = [v],$$

which holds precisely when v is an eigenvector of A . Thus the fixed points of φ are exactly the eigenlines of A .

The characteristic polynomial of A has degree two and splits over $\mathbb{C}$. If it has two distinct roots, then A has exactly two eigenlines. If it has one repeated root, then either its eigenspace is one-dimensional, giving one eigenline, or its eigenspace is two-dimensional, in which case A is scalar. The latter would make φ the identity, contrary to the hypothesis. Therefore φ has exactly one or two fixed points. $\square$

9 Harmonic Functions

9.1 Harmonicity and Holomorphic Functions

Definition 9.1

$G \subseteq \mathbb{C} \cong \mathbb{R}^2$ open, $f : G \rightarrow \mathbb{C}$ is *harmonic* if it is in $\mathcal{C}^2$ (which means that the second order partials are continuous) and

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0 \text{ (Laplacian Equation)}$$

Note

f is harmonic if and only if $u = \operatorname{Re}(f)$ and $v = \operatorname{Im}(f)$ are harmonic.

Example

$f(x, y) = x^2 - y^2$, $g(x, y) = 2xy$ are harmonic functions.

Proposition 9.1

If $f : G \rightarrow \mathbb{C}$ is holomorphic, then it is harmonic.

Proof: Write $f = u + iv$, where $u = \operatorname{Re}(f)$ and $v = \operatorname{Im}(f)$. Since holomorphic functions are smooth, $u, v \in \mathcal{C}^2$. The Cauchy-Riemann equations give

$$u_x = v_y \quad \text{and} \quad u_y = -v_x.$$

Differentiating the first equation with respect to x and the second with respect to y gives

$$u_{xx} = v_{yx} \quad \text{and} \quad u_{yy} = -v_{xy}.$$

By Clairaut's theorem, the mixed partials of v are equal, so

$$u_{xx} + u_{yy} = v_{yx} - v_{xy} = 0.$$

Thus u is harmonic.

Similarly, differentiating $u_x = v_y$ with respect to y and $u_y = -v_x$ with respect to x gives

$$v_{yy} = u_{xy} \quad \text{and} \quad v_{xx} = -u_{yx}.$$

By Clairaut's theorem, the mixed partials of u are equal, so

$$v_{xx} + v_{yy} = -u_{yx} + u_{xy} = 0.$$

Therefore v is harmonic, and hence f is harmonic. □

9.2 Harmonic Conjugates

Definition 9.2

Let $u : G \rightarrow \mathbb{R}$ be harmonic. Then $v : G \rightarrow \mathbb{R}$ is a *harmonic conjugate* if $f(x + iy) = u(x, y) + iv(x, y)$ is holomorphic.

Theorem 9.2

If G is simply connected, $u : G \rightarrow \mathbb{R}$ is harmonic, then u has a harmonic conjugate.

Proof: Define

$$f = u_x - iu_y.$$

Write $f = p + iq$, where $p = u_x$ and $q = -u_y$. Since u is harmonic,

$$p_x = u_{xx} = -u_{yy} = q_y.$$

Moreover, equality of mixed partials gives

$$q_x = -u_{yx} = -u_{xy} = -p_y.$$

Thus f satisfies the Cauchy-Riemann equations, so it is holomorphic.

By Theorem 7.19 (Characterizations of Simply Connected Regions), f has a primitive on G . Hence there is a holomorphic function $g : G \rightarrow \mathbb{C}$ such that $g' = f$. Fix $z_0 \in G$. After adding a constant to g , we may assume that $g(z_0) = u(z_0)$. Write

$$g = u_1 + iv.$$

Since g is holomorphic, the Cauchy-Riemann equations imply

$$g' = (u_1)_x - i(u_1)_y.$$

Comparing this with

$$g' = f = u_x - iu_y$$

gives

$$(u_1)_x = u_x \quad \text{and} \quad (u_1)_y = u_y.$$

Therefore $u_1 - u$ has zero gradient on the connected set G , so it is constant. Since

$$(u_1 - u)(z_0) = \operatorname{Re}(g(z_0)) - u(z_0) = 0,$$

we have $u_1 = u$. Consequently, $g = u + iv$ is holomorphic, and hence v is a harmonic conjugate of u . □

Example

Let $u(x, y) = \frac{1}{2} \log(x^2 + y^2)$. Define for $z = x + iy$, $0 < |z| < R$. It is easy to verify that u is harmonic.

However, $\frac{1}{z}$ does not admit an anti-derivative on a punctured disk G .

Simply connected ==> connected?

Corollary 9.3

Let $G \subseteq \mathbb{C}$ be open and connected. Then G is simply-connected if and only if every harmonic function on G admits a harmonic conjugate.

Proof: ($\Rightarrow$) Theorem from before.

($\Leftarrow$) Transport around the last example (fill in the details). □

Corollary 9.4

Identity Theorem

G is open and connected, and harmonic functions u and v agree on a non-empty open subset of G , then $u = v$.

Proof: We may assume $v = 0$ and u is $\mathbb{R}$ -valued. Let $H = \operatorname{int}(u^{-1}(\{0\}))$, which by assumption, is non-empty. □

Remark

If $u = v$ on any subset admitting a cluster point in G , then $u = v$ on G .